Integrator and Tolerance Bench on Small Mode Networks

21 October 2026

In short

No run is singled out as a reference here; every statement compares runs with each other. Δ means the detuning throughout.

1. **The absolute tolerance, not the integrator, causes the disagreements seen in earlier sweeps.**
   - At atol $= 10^{-40}$ (the sweep default so far), bursts happen at different times for different integrators on four of the nine networks (and on `2p_M2.0T_eth`, see item 5).
   - At atol $= 10^{-300}$, the five integrators give the same curves on eight of the nine networks.
2. atol $= 0$ **(pure relative error control) reproduces** atol $= 10^{-300}$ **exactly wherever it finishes.** It does not finish on `2p_M2.0T_eth`, where an amplitude decays below the smallest normal double: rk8pd crawls into the wall-time limit there, and CVODE stops with "too much accuracy requested".
3. **rk8pd is the cheapest integrator.** At atol $= 10^{-300}$, measured against rk8pd (medians over the networks):
   - BDF takes 11.6× the steps and 5.8× the wall time;
   - Adams takes 5.2× the steps and 2.9× the wall time.
4. **BDF-Krylov (GMRES with a banded finite-difference preconditioner) gives the same curves as BDF with the analytic Jacobian, but it is not faster at these sizes (≤ 6 modes).** Preconditioner band 3 and the full band behave the same.
5. **`2p_M2.0T_eth` disagrees between runs at every setting.** One daughter's exact dip ($E \sim 10^{-3600}$) is far below what a double can hold.
6. **Proposal for the sweep:** rk8pd, rtol $= 10^{-8}$, atol $= 10^{-300}$. The decisions are listed in §**??**.

1 Setup

1.1 Equations and notation

Each mode a has a slow complex amplitude $A_a(t)$, with t in seconds. For a triplet (a, b, c) with $b \neq c$, mode a obeys

$$\dot{A}_a = -\gamma_a A_a + 2i\,\omega_a\,\kappa_{abc}\,A_b^* A_c^*\,e^{i\Delta_{abc}t}, \qquad \Delta_{abc} = \omega_a + \omega_b + \omega_c. \tag{1}$$

Each triplet that contains a adds such a term. A self-coupled pair ($b = c$) has the coefficient 1 instead of 2 in the parent's equation. The symbols are:

- ω_a: the mode frequency, signed by generation. Generation 0 (the parents) has $-|\omega|$; generation 1 (the daughters) has $+|\omega|$.
- Δ_{abc}: the detuning of the triplet. With the signs above it is small for a parent → daughter + daughter triplet. It is fixed by the network: it is the same in every run of a given network.
- γ_a: the linear damping rate, in s^{-1}. $\gamma_a < 0$ means the mode is driven (the parents); $\gamma_a > 0$ means it is damped.
- κ_{abc}: the three-mode coupling coefficient.
- $E_a = |A_a|^2$: the mode energy, in units of $E_\star$.
- E_{th}: a parent's parametric threshold energy for its selecting pair (Weinberg+12, eq. 50).
- E_{eq}: the three-mode equilibrium energy (MW25, eq. 6). It is absent when the triplet has no stable equilibrium.

The integrators advance the real vector $y = (\mathrm{Re}\,A_0, \mathrm{Im}\,A_0, \ldots, \mathrm{Re}\,A_{n-1}, \mathrm{Im}\,A_{n-1})$ of length $2n$, where n is the number of modes.

1.2 Error control and the energy floor

The step test. Both libraries accept a step when the local error estimate e_i of each component y_i satisfies

$$|e_i| \lesssim \text{atol} + \text{rtol}\,|y_i|. \tag{2}$$

GSL applies this test to each component separately. CVODE applies it through a weighted root-mean-square norm, with weights $1/(\text{rtol}|y_i| + \text{atol})$.

The floor.

- When $|y_i| \gg \text{atol/rtol}$, the error control is relative. Below that, it is absolute: the component is held only to about $\pm$atol.
- So a decaying mode is not followed below $|A| \sim \text{atol}$, i.e. $E \sim \text{atol}^2$. Each integrator leaves the mode at its own level near this floor.
- When the mode later grows back at rate σ, it needs a time of about $\ln(E_{\text{th}}/E_{\text{floor}})/(2\sigma)$ to reach threshold. A floor that differs by tens of decades therefore moves the next burst.

Why atol $= 0$ **is fragile.**

- atol $= 0$ removes the floor but asks for relative accuracy at every magnitude.
- Doubles below 2.2×10^{-308} are subnormal and carry fewer significant digits. Once an amplitude reaches that range ($E \lesssim 5 \times 10^{-616}$), the requested accuracy cannot be met: rk8pd's step size collapses, and CVODE stops.
- A mode decaying freely from $E = 10^{-18}$ reaches that range after about $690/\gamma$.

Table 1: The absolute tolerances tested and the energy floor each one implies.

atol	floor $E \sim \text{atol}^2$	note
10^{-40}	10^{-80}	sweep default so far
10^{-150}	10^{-300}	
10^{-300}	10^{-600}	near the double-precision limit for A
0	none	pure relative control; fails once an A becomes subnormal ($E \lesssim 5 \times 10^{-616}$)

1.3 Integrators

Table 2: The five integrator configurations. GSL 2.8; CVODE from SUNDIALS 7.9.0 (the cluster has 7.1.1).

label	method	linear algebra
rk8pd	GSL `rk8pd`: explicit embedded Runge–Kutta Prince–Dormand 8(9)	none
BDF, dense analytic Jacobian	CVODE BDF, orders 1–5, Newton iteration, stability-limit detection	dense LU on the hand-written Jacobian
BDF-Krylov, band full	as BDF	GMRES (SPGMR, Krylov dimension 5), with Jacobian–vector products by finite differences. Left preconditioner CVBANDPRE: a banded finite-difference Jacobian of half-bandwidth $2n - 1$ (the whole matrix)
BDF-Krylov, band 3	as BDF	as above, half-bandwidth 3
Adams, fixed point	CVODE Adams–Moulton, orders 1–12, fixed-point iteration	none

Restarts.

- CVODE is re-initialised (`CVodeReInit`) at each of the 4000 outputs, with its local time set back to 0. GSL likewise restarts its local time in each output interval.
- The phase factor $e^{i\Delta t}$ at the restart time is carried along as a constant.
- This keeps t small inside the exponential. Otherwise rounding of $t \approx 10^{13}$ s would add phase noise of order $\Delta\,\varepsilon\,t$, where ε is the machine epsilon.
- The restarts cost almost nothing. BDF at atol $= 10^{-40}$ on `1p2d_M1.7_eth` took 6.39 M steps with them, against 6.92 M steps in the earlier cluster run without them.

1.4 Runs

- **Networks.** Nine networks (Table **??**), made by `network_build` with $N = 1$ or 4, cut 0.15 and closure 0.005.

  - The name prefix gives the shape: `1p2d` is one parent with one daughter pair, `1p4d` is one parent with two pairs, and `2p` is two parents with one pair each.
  - The suffix gives the pair-ranking rule.
- **Initial state.** Parents start at $|A| = 10^{-6}$ ($E = 10^{-12}$) and daughters at $|A| = 10^{-9}$ ($E = 10^{-18}$). The phases are random, with seed 1.
- **Time span.** $t_{\text{end}} = 10^{13}$ s, with 4000 equally spaced outputs. A run stops early, as a runaway, once $\sum_a E_a > 10^{-2}$.
- **Tolerances.**
  - All five integrators: rtol $= 10^{-8}$, with atol $\in \{10^{-40}, 10^{-150}, 10^{-300}, 0\}$.
  - rk8pd also ran at rtol $= 10^{-10}$ (atol $= 0$ and 10^{-40} only) and at rtol $= 10^{-12}$ (all four atol); see Appendix **??**.
- **Wall-time limits.**
  - `network_run -max-seconds 900`. It is checked only between outputs, so it cannot stop a run that stalls inside one output interval.
  - The bench scripts also kill a run after 600 s (rk8pd) or 1000 s (CVODE).
- **Machine.** A laptop running up to six runs at once. Wall times are good to about a factor of 2; step counts are exact.

Stop codes.

- `end`: reached t_{end}.
- `runaway`: crossed $\sum E = 10^{-2}$.
- `failed`: CVODE returned an error. All four failures here are "too much accuracy requested".
- `wall`: stopped by the 900 s limit.
- `stalled`: killed by the bench script's timeout; no output was written.

1.5 Table columns

The integrator advances in time steps whose length h it picks itself. The basic unit of work is one evaluation of the right-hand side $f(t, y) = \dot{y}$, i.e. $\dot{A}$ for every mode. The examples are `1p2d_M1.7_eth` at atol $= 10^{-300}$ (§**??**). "–" means the column does not apply to that integrator.

steps Accepted steps: how many times the solution was advanced in time.

- rk8pd took 1.13 M.
- BDF took 13.78 M. It is lower order, so it needs shorter steps for the same accuracy.

rejected
Steps that were tried, thrown away and retried with a shorter step. They cost work but do not advance the solution.

- rk8pd rejects a step when its error estimate exceeds atol + rtol$|y_i|$.
- CVODE also rejects a step when the iteration that solves for the new state (see NL it.) fails to converge.

RHS Right-hand-side evaluations, counted inside our f, so everything that calls it is included. This is the fairest cost measure that does not depend on the machine.

- **rk8pd:** about 12 per attempted step (13 stages, the first one reused). Here $14.65\,\text{M} \approx 12 \times (1.13\,\text{M} + 0.13\,\text{M})$.
- **BDF and Adams:** one per nonlinear iteration, so RHS ≈ NL it.
- **BDF-Krylov:** one per nonlinear iteration, plus one per GMRES iteration, plus about $2n$ per preconditioner build. Here $29.86\,\text{M} \approx 14.47\,\text{M} + 13.98\,\text{M} + 6 \times 234.6\,\text{k}$, which is about twice dense BDF's count for the same number of steps.

Jac. Jacobian evaluations. The Jacobian is the $2n \times 2n$ matrix $J = \partial f / \partial y$.

- **BDF:** computed from the hand-written formula. CVODE reuses J across many steps and recomputes it only when its iteration starts to converge slowly: 241 k Jacobians for 13.78 M steps.
- **BDF-Krylov:** CVBANDPRE's banded finite-difference approximation of J, used as the preconditioner.
- **rk8pd and Adams:** use no Jacobian.

NL it.
Nonlinear iterations. BDF and Adams are implicit: the new state appears on both sides of the update,

$$y_{\text{new}} = (\text{terms from earlier steps}) + h\,\beta\,f(t_{\text{new}}, y_{\text{new}}),$$

with β a fixed coefficient of the method. The integrator therefore solves for y_{new} iteratively.

- **BDF:** Newton iteration, which uses J.
- **Adams:** fixed-point iteration (put the current guess into f and repeat), which needs no J.
- Here it takes about one iteration per step: 14.74 M iterations for 13.78 M BDF steps.
- **rk8pd:** explicit, so it has none.

lin. it.

GMRES iterations; BDF-Krylov only. Each Newton iteration solves a linear system with the matrix $I - h\beta J$.

- **Dense BDF:** solves it directly by LU factorisation, so there are no iterations to count.
- **BDF-Krylov:** solves it iteratively with GMRES. Each GMRES iteration needs one product Jv, which is approximated with one extra right-hand-side evaluation.

wall [s]

Elapsed clock time of the integration, in seconds. Up to six runs shared the laptop, so it is good only to about a factor of 2. Use steps and RHS for exact comparisons.

2 Overview

Table 3: The networks. Each name links to its page.

network	model	modes	parents	triplets	rule
`1p2d_M1.7_eth`	`dsct_M1.7_T7750`	3	1	1	eth
`1p2d_M2.0T_eth`	`dsct_M2.0_T7696`	3	1	1	eth
`1p2d_M2.0_dgamma`	`dsct_M2.0`	3	1	1	dgamma
`1p4d_M1.7_eth`	`dsct_M1.7_T7750`	5	1	2	eth
`1p4d_M2.0T_eth`	`dsct_M2.0_T7696`	5	1	2	eth
`1p4d_M2.0_eth`	`dsct_M2.0`	5	1	2	eth
`2p_M1.7_eth`	`dsct_M1.7_T7750`	6	2	2	eth
`2p_M2.0T_eth`	`dsct_M2.0_T7696`	5	2	2	eth
`2p_M2.0_eeq`	`dsct_M2.0`	6	2	2	eeq

Table 4: Cost of each (integrator, atol) pair over the nine networks at rtol $= 10^{-8}$. The ratios are medians over the networks where both this run and rk8pd at the same atol finished. * Sum over the finished runs only.

integrator	atol	stopped early (of 9)	steps / rk8pd median	wall / rk8pd median	$\sum$ steps	$\sum$ wall [s]
rk8pd	10^{-40}	0	1.00	1.00	26.03 M	58
	10^{-150}	0	1.00	1.00	29.99 M	99
	10^{-300}	0	1.00	1.00	32.16 M	85
	0	1	1.00	1.00	31.77 M*	94*
BDF, dense analytic Jacobian	10^{-40}	0	8.71	4.44	155.79 M	255
	10^{-150}	0	11.60	4.21	359.03 M	480
	10^{-300}	0	11.59	5.81	372.31 M	530
	0	1	11.91	4.31	368.85 M*	528*
BDF-Krylov, band full	10^{-40}	0	9.22	13.10	314.58 M	569
	10^{-150}	0	11.30	7.07	347.45 M	633
	10^{-300}	0	11.30	7.62	359.96 M	585
	0	1	11.62	6.39	356.55 M*	639*
BDF-Krylov, band 3	10^{-40}	0	9.20	11.31	304.81 M	557
	10^{-150}	0	10.98	7.16	337.80 M	701
	10^{-300}	0	10.98	8.51	350.54 M	744
	0	1	11.23	7.05	347.22 M*	699*
Adams, fixed point	10^{-40}	0	5.64	4.14	157.31 M	208
	10^{-150}	0	5.28	2.45	165.80 M	196
	10^{-300}	0	5.21	2.88	168.85 M	202
	0	1	5.21	1.86	166.85 M*	167*

2.1 Findings

1. **atol, not the integrator, sets the curves.**

- At atol $= 10^{-300}$ and at atol $= 0$, the five integrators agree to within 0.25 decades in E at every output where $E > 10^{-34}$ (the plots' lower edge) on eight networks; the exception is `2p_M2.0T_eth`. By eye, their plots are the same.
- At atol $= 10^{-40}$, bursts move on four networks: `1p2d_M1.7_eth`, `1p2d_M2.0_dgamma`, `2p_M1.7_eth` and `2p_M2.0_eeq`.

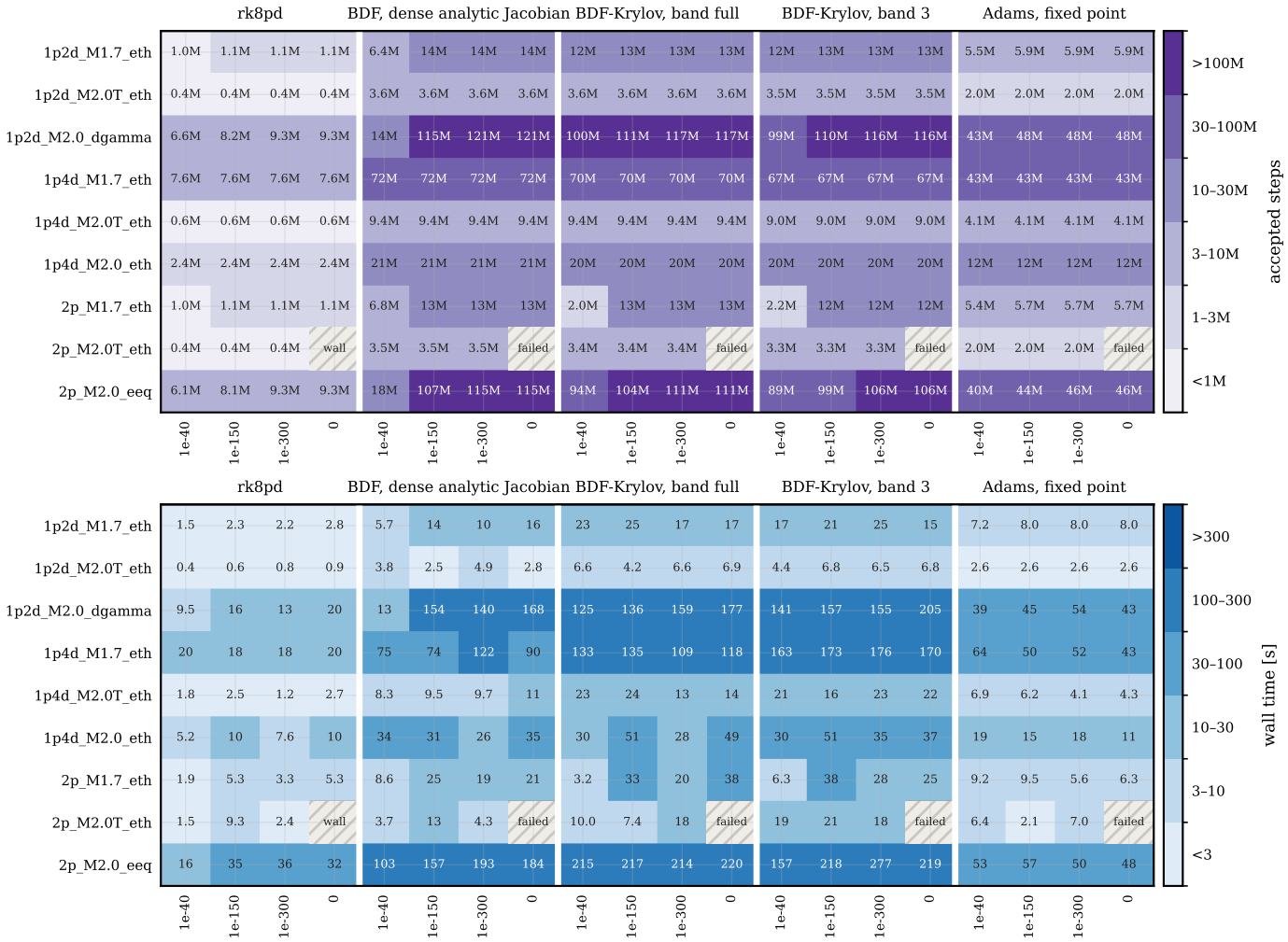


Figure 1: Cost of every network × (integrator, atol) at rtol = 10^{-8}. **Top:** accepted steps. **Bottom:** wall time. Hatched cells stopped early (`failed`, `wall` or `stalled`).

  - The bursts move between integrators.
  - Each integrator's bursts also move relative to its own atol $= 10^{-300}$ run.
  - The cause is that each integrator leaves the decaying daughters at a different level around the 10^{-80} floor, and the next burst follows from that level.
- This explains the earlier rk8pd–BDF disagreement on `1p2d_M1.7_eth`: both were run at atol $= 10^{-40}$.
- atol $= 10^{-150}$ removes the differences on the two M1.7 networks, but not on `dgamma` and `eeq`.

2. atol $= 10^{-300}$ **and** atol $= 0$ **give identical curves for every integrator, wherever** atol $= 0$ **finishes.** atol $= 0$ does not finish on `2p_M2.0T_eth`:
   - rk8pd crawls to the wall-time limit at 5.1×10^{11} s, after 50 M steps;
   - all CVODE configurations fail at 4.85×10^{11} s.
3. **rk8pd is the cheapest integrator.** At every atol it takes the fewest steps on every network. It is also the fastest by wall time, with one exception: Adams on `2p_M2.0T_eth` at 10^{-150}, which is within the timing noise.
4. **BDF-Krylov gives no gain at these sizes.**
   - The finite-difference Jacobian is as good as the hand-written one: the curves are the same, and BDF-Krylov takes 1–8 % fewer steps than dense BDF.
   - Each step costs more. The median wall time is 7.6–8.5× rk8pd's, against 5.8× for dense BDF.
   - Band 3 and the full band behave the same.
   - With $2n \leq 12$ unknowns, a dense LU solve is trivial. A Krylov solver is expected to pay off only for hundreds of modes; this was not tested.
5. rtol $= 10^{-8}$ **is enough for rk8pd** (Appendix **??**). At atol $= 10^{-300}$, rk8pd at rtol $= 10^{-8}$ and at rtol $= 10^{-12}$ agree to within 0.01 decades on every network except `2p_M2.0T_eth`.
6. `2p_M2.0T_eth` **cannot be converged in double precision.**
   - The daughter (14,−48) is fed by parent (2,−3) through a self-coupled pair. In exact arithmetic it decays to $E \sim 10^{-3600}$ by 3×10^{12} s.
   - Its regrowth time is set by whichever floor a run has. So the runs disagree with each other at every atol, at every rtol, and for every integrator.
   - Any network whose exact dips go below about 10^{-600} will behave like this.

3 Decisions

1. **Sweep integrator and tolerances.**
   (a) *rk8pd,* rtol $= 10^{-8}$*,* atol $= 10^{-300}$ *(proposed).*
   - The same curves as every other integrator and as atol $= 0$.
   - The cheapest option, and it never stalled.
   - Requires changing `RUN_ARGS` in `network_sweep.sh` from `-atol 1e-40`.

   (b) atol $= 0$. The same curves, but it fails on networks with very deep dips. A stalled run also defeats `-max-seconds`.
   (c) *A physical floor,* atol $= \sqrt{E_{\min}}$ (the "$E_{\min,\mathrm{crash}}$" idea).
   - This is a choice of physics, not of numerics.
   - The `dgamma` and `eeq` networks show that a floor of $E_{\min} \gtrsim 10^{-300}$ already moves bursts.
   - Setting a mode to zero is not an option: a pair whose daughters are both zero can never regrow.
2. **Role of CVODE.** Keep it as an independent check on a sample of networks. Adams is the cheaper choice (5× rk8pd's steps, 3× its wall time); BDF is the established one.
3. **BDF-Krylov:** keep it (about ten lines; `-solver bdf-krylov -band`) for future large networks, or remove it.
4. **Floor-dependent networks such as `2p_M2.0T_eth`.** Three options:
   (a) Flag runs in which a mode stays below 10^{-600}, and treat their timing as unconverged.
   (b) Pick a physical noise floor, and accept that results depend on it.
   (c) Integrate $\ln|A|$ and the phase instead of $(\mathrm{Re}\,A, \mathrm{Im}\,A)$. This is exact at any depth, but it is the largest change.
5. **Reference for future comparisons:** still to be chosen.
6. **Open item:** `-max-seconds` does not fire inside a stalled output interval.

Caveats.

- One random seed.

- At most six modes per network.
- CVODE was run only at rtol $= 10^{-8}$.
- Wall times are noisy, because the CPU was shared.

4 Networks

Each network has one page.

- **Header.**
  - The modes: id, generation, (ℓ, n), ω and γ.
  - The selecting pairs, with detuning Δ/ω_a, κ, E_{th} and E_{eq}.
- **Plots.** The `network_plot.py` plot of each run, unchanged. Rows are integrators and columns are atol, all at rtol $= 10^{-8}$.
  - Each plot keeps its own time axis, so a run that stopped early covers a shorter span.
  - The dashed lines are each parent's E_{th} and E_{eq}.
  - A framed box stands in for a run with no output.
  - The images are high resolution, so zoom in for detail.
- **Table.** One row per run; the columns are defined in §**??**.

4.1 `1p2d_M1.7_eth`

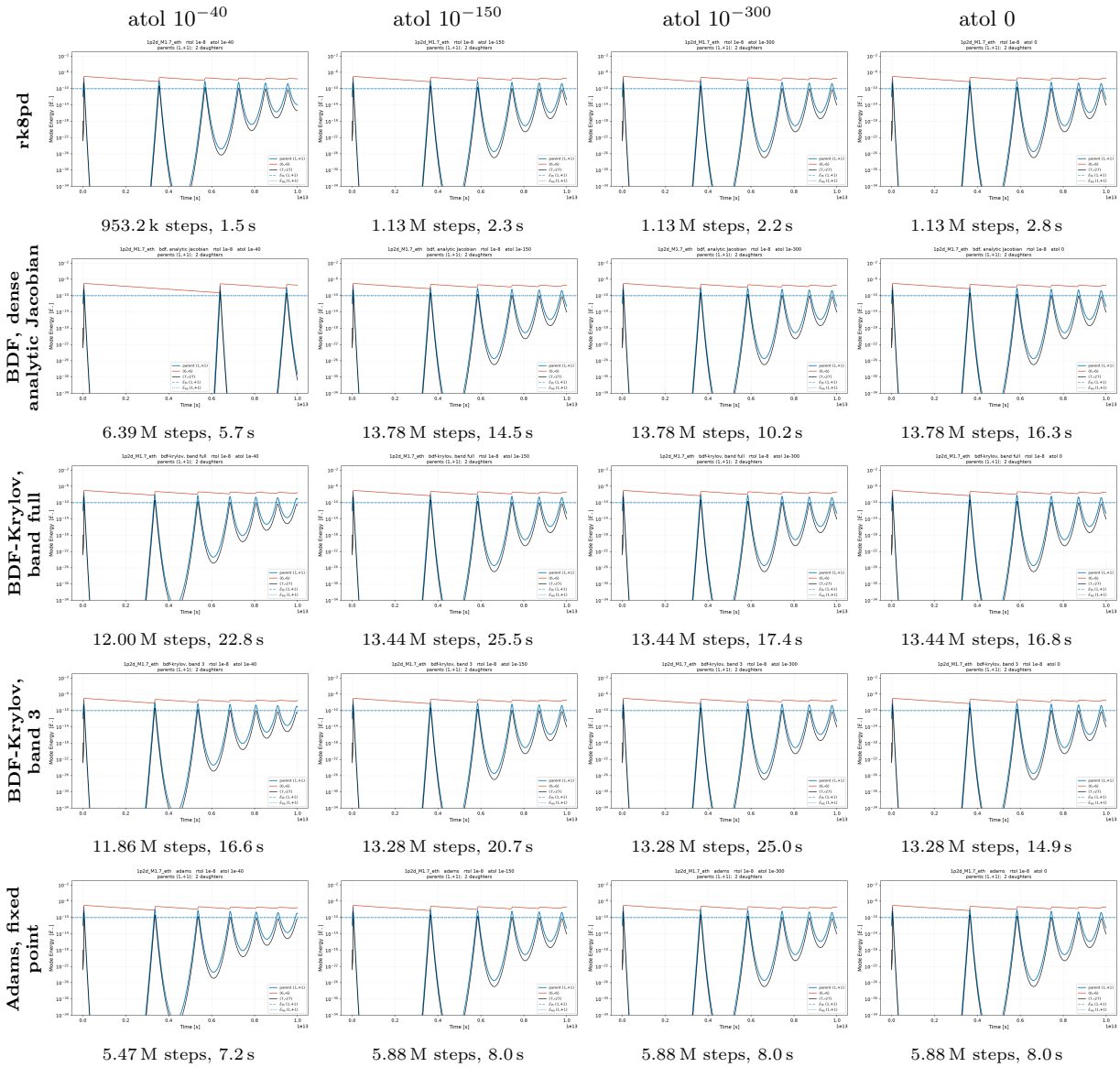
id	gen	(ℓ, n)	ω [rad s^{-1}]	γ [s^{-1}]
0	0	(1,+1)	1.402×10^{-3}	-1.150×10^{-10}
1	1	(6,−6)	1.070×10^{-3}	4.184×10^{-13}
2	1	(7,−27)	3.326×10^{-4}	8.612×10^{-10}

Model `dsct_M1.7_T7750`, rule eth, $N = 1$, cut 0.15, closure 0.005; 3 modes, 1 triplet(s).

Selecting pairs

(1,+1) → (6,−6) + (7,−27): $\Delta/\omega_a = -3.98 \times 10^{-5}$, $\kappa = +0.107$, $E_{\rm th} = 9.22 \times 10^{-11}$, $E_{\rm eq} = 1.23 \times 10^{-10}$

The second burst disagreed between rk8pd and BDF in the earlier sweep.



integrator	atol	stop	$t_{\rm stop}$ [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
rk8pd	10^{-40}	end	1.00×10^{13}	953.2 k	101.3 k	12.39 M	–	–	–	1.5
	10^{-150}	end	1.00×10^{13}	1.13 M	126.8 k	14.65 M	–	–	–	2.3
	10^{-300}	end	1.00×10^{13}	1.13 M	126.8 k	14.65 M	–	–	–	2.2
	0	end	1.00×10^{13}	1.13 M	126.8 k	14.65 M	–	–	–	2.8
BDF, dense analytic Jacobian	10^{-40}	end	1.00×10^{13}	6.39 M	49.0 k	6.85 M	117.8 k	6.85 M	–	5.7
	10^{-150}	end	1.00×10^{13}	13.78 M	118.0 k	14.74 M	241.1 k	14.74 M	–	14.5
	10^{-300}	end	1.00×10^{13}	13.78 M	118.0 k	14.74 M	241.1 k	14.74 M	–	10.2
	0	end	1.00×10^{13}	13.78 M	118.0 k	14.74 M	241.1 k	14.74 M	–	16.3
BDF-Krylov, band full	10^{-40}	end	1.00×10^{13}	12.00 M	104.9 k	27.04 M	209.5 k	12.90 M	12.88 M	22.8
	10^{-150}	end	1.00×10^{13}	13.44 M	125.8 k	29.86 M	234.6 k	14.47 M	13.98 M	25.5
	10^{-300}	end	1.00×10^{13}	13.44 M	125.8 k	29.86 M	234.6 k	14.47 M	13.98 M	17.4
	0	end	1.00×10^{13}	13.44 M	125.8 k	29.86 M	234.6 k	14.47 M	13.98 M	16.8
BDF-Krylov, band 3	10^{-40}	end	1.00×10^{13}	11.86 M	110.2 k	26.98 M	207.1 k	12.80 M	12.94 M	16.6
	10^{-150}	end	1.00×10^{13}	13.28 M	132.1 k	29.80 M	231.8 k	14.35 M	14.05 M	20.7
	10^{-300}	end	1.00×10^{13}	13.28 M	132.1 k	29.80 M	231.8 k	14.35 M	14.05 M	25.0
	0	end	1.00×10^{13}	13.28 M	132.1 k	29.80 M	231.8 k	14.35 M	14.05 M	14.9
Adams, fixed point	10^{-40}	end	1.00×10^{13}	5.47 M	231.7 k	8.22 M	–	8.22 M	–	7.2
	10^{-150}	end	1.00×10^{13}	5.88 M	285.4 k	8.57 M	–	8.57 M	–	8.0
	10^{-300}	end	1.00×10^{13}	5.88 M	285.4 k	8.57 M	–	8.57 M	–	8.0
	0	end	1.00×10^{13}	5.88 M	285.4 k	8.57 M	–	8.57 M	–	8.0

4.2 `1p2d_M2.0T_eth`

id	gen	(ℓ, n)	ω [rad s^{-1}]	γ [s^{-1}]
0	0	(2,−2)	7.959×10^{-4}	-6.697×10^{-11}
1	1	(3,−31)	1.287×10^{-4}	8.096×10^{-11}
2	1	(5,−8)	6.672×10^{-4}	7.367×10^{-13}

Model `dsct_M2.0_T7696`, rule eth, $N = 1$, cut 0.15, closure 0.005; 3 modes, 1 triplet(s).

Selecting pairs
(2,−2) → (3,−31) + (5,−8): $\Delta/\omega_a = -3.69 \times 10^{-6}$, $\kappa = +0.215$, $E_{\text{th}} = 4.87 \times 10^{-12}$, $E_{\text{eq}} = 1.50 \times 10^{-10}$

integrator	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
rk8pd	10^{-40}	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	0.4
	10^{-150}	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	0.6
	10^{-300}	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	0.8
	0	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	0.9
BDF, dense analytic Jacobian	10^{-40}	end	1.00×10^{13}	3.60 M	59.0 k	3.99 M	77.3 k	3.99 M	–	3.8
	10^{-150}	end	1.00×10^{13}	3.60 M	59.0 k	3.99 M	77.3 k	3.99 M	–	2.5
	10^{-300}	end	1.00×10^{13}	3.60 M	59.0 k	3.99 M	77.3 k	3.99 M	–	4.9
	0	end	1.00×10^{13}	3.60 M	59.0 k	3.99 M	77.3 k	3.99 M	–	2.8
BDF-Krylov, band full	10^{-40}	end	1.00×10^{13}	3.55 M	60.9 k	7.98 M	73.4 k	3.91 M	3.62 M	6.6
	10^{-150}	end	1.00×10^{13}	3.55 M	60.9 k	7.98 M	73.4 k	3.91 M	3.62 M	4.2
	10^{-300}	end	1.00×10^{13}	3.55 M	60.9 k	7.98 M	73.4 k	3.91 M	3.62 M	6.6
	0	end	1.00×10^{13}	3.55 M	60.9 k	7.98 M	73.4 k	3.91 M	3.62 M	6.9
BDF-Krylov, band 3	10^{-40}	end	1.00×10^{13}	3.54 M	61.1 k	7.98 M	73.2 k	3.91 M	3.63 M	4.4
	10^{-150}	end	1.00×10^{13}	3.54 M	61.1 k	7.98 M	73.2 k	3.91 M	3.63 M	6.8
	10^{-300}	end	1.00×10^{13}	3.54 M	61.1 k	7.98 M	73.2 k	3.91 M	3.63 M	6.5
	0	end	1.00×10^{13}	3.54 M	61.1 k	7.98 M	73.2 k	3.91 M	3.63 M	6.8
Adams, fixed point	10^{-40}	end	1.00×10^{13}	2.04 M	134.1 k	3.35 M	–	3.34 M	–	2.6
	10^{-150}	end	1.00×10^{13}	2.04 M	134.1 k	3.35 M	–	3.34 M	–	2.6
	10^{-300}	end	1.00×10^{13}	2.04 M	134.1 k	3.35 M	–	3.34 M	–	2.6
	0	end	1.00×10^{13}	2.04 M	134.1 k	3.35 M	–	3.34 M	–	2.6

4.3 `1p2d_M2.0_dgamma`

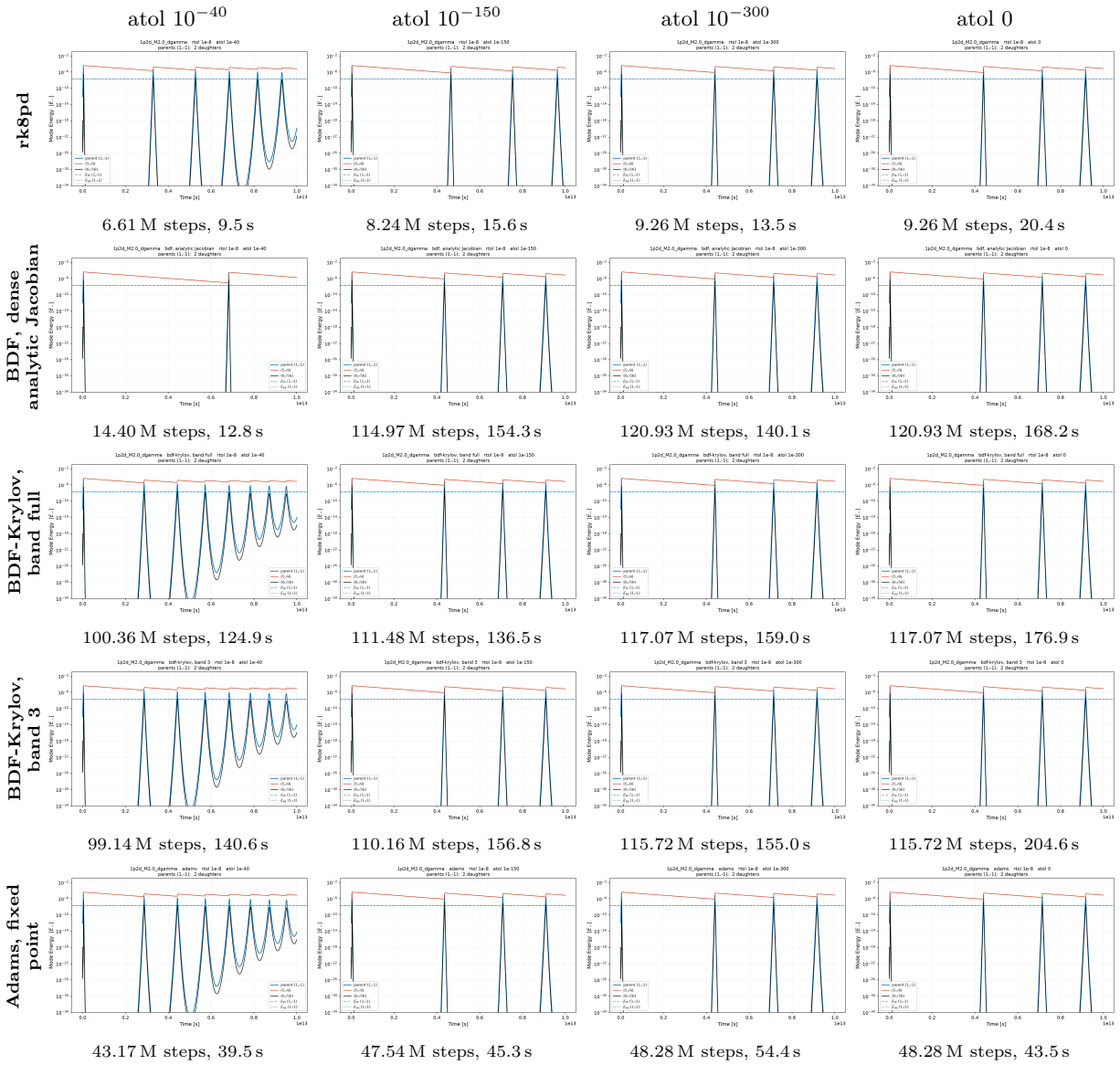
id	gen	(ℓ, n)	ω [rad s^{-1}]	γ [s^{-1}]
0	0	(1,−1)	7.581×10^{-4}	-2.276×10^{-10}
1	1	(5,−9)	6.198×10^{-4}	4.584×10^{-13}
2	1	(6,−56)	1.377×10^{-4}	2.817×10^{-9}

Model `dsct_M2.0`, rule dgamma, $N = 1$, cut 0.15, closure 0.005; 3 modes, 1 triplet(s).

Selecting pairs

(1,−1) → (5,−9) + (6,−56): $\Delta/\omega_a = -6.83 \times 10^{-4}$, $\kappa = +0.085$, $E_{\rm th} = 1.75 \times 10^{-8}$, $E_{\rm eq} = 2.07 \times 10^{-8}$

Deep dips between bursts.



integrator	atol	stop	$t_{\rm stop}$ [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
rk8pd	10^{-40}	end	1.00×10^{13}	6.61 M	575.7 k	85.89 M	–	–	–	9.5
	10^{-150}	end	1.00×10^{13}	8.24 M	937.9 k	107.11 M	–	–	–	15.6
	10^{-300}	end	1.00×10^{13}	9.26 M	1.04 M	120.43 M	–	–	–	13.5
	0	end	1.00×10^{13}	9.26 M	1.04 M	120.43 M	–	–	–	20.4
BDF, dense analytic Jacobian	10^{-40}	end	1.00×10^{13}	14.40 M	215.6 k	16.26 M	378.7 k	16.26 M	–	12.8
	10^{-150}	end	1.00×10^{13}	114.97 M	655.1 k	121.60 M	1.93 M	121.59 M	–	154.3
	10^{-300}	end	1.00×10^{13}	120.93 M	719.2 k	127.92 M	2.03 M	127.91 M	–	140.1
	0	end	1.00×10^{13}	120.93 M	719.2 k	127.92 M	2.03 M	127.91 M	–	168.2
BDF-Krylov, band full	10^{-40}	end	1.00×10^{13}	100.36 M	648.8 k	225.12 M	1.69 M	107.09 M	107.90 M	124.9
	10^{-150}	end	1.00×10^{13}	111.48 M	738.1 k	246.46 M	1.87 M	118.93 M	116.29 M	136.5
	10^{-300}	end	1.00×10^{13}	117.07 M	821.6 k	257.00 M	1.97 M	124.94 M	120.27 M	159.0
	0	end	1.00×10^{13}	117.07 M	821.6 k	257.00 M	1.97 M	124.94 M	120.27 M	176.9
BDF-Krylov, band 3	10^{-40}	end	1.00×10^{13}	99.14 M	679.3 k	223.80 M	1.66 M	106.16 M	107.65 M	140.6
	10^{-150}	end	1.00×10^{13}	110.16 M	777.2 k	244.85 M	1.85 M	117.94 M	115.81 M	156.8
	10^{-300}	end	1.00×10^{13}	115.72 M	853.2 k	255.64 M	1.94 M	123.92 M	120.06 M	155.0
	0	end	1.00×10^{13}	115.72 M	853.2 k	255.64 M	1.94 M	123.92 M	120.06 M	204.6
Adams, fixed point	10^{-40}	end	1.00×10^{13}	43.17 M	1.65 M	61.88 M	–	61.88 M	–	39.5
	10^{-150}	end	1.00×10^{13}	47.54 M	2.13 M	69.35 M	–	69.35 M	–	45.3
	10^{-300}	end	1.00×10^{13}	48.28 M	2.32 M	68.68 M	–	68.68 M	–	54.4
	0	end	1.00×10^{13}	48.28 M	2.32 M	68.68 M	–	68.68 M	–	43.5

4.4 1p4d_M1.7_eth

id	gen	(ℓ, n)	ω [rad s^{-1}]	γ [s^{-1}]
0	0	(1,+1)	1.402×10^{-3}	-1.150×10^{-10}
1	1	(6,−6)	1.070×10^{-3}	4.184×10^{-13}
2	1	(7,−27)	3.326×10^{-4}	8.612×10^{-10}
3	1	(5,−14)	4.541×10^{-4}	7.998×10^{-12}
4	1	(6,−7)	9.483×10^{-4}	7.855×10^{-13}

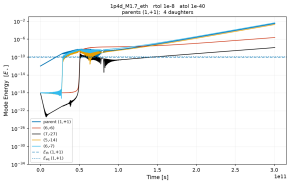
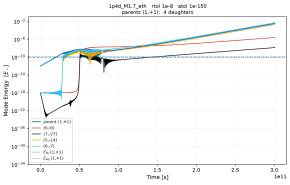
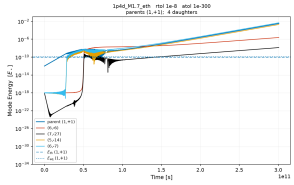
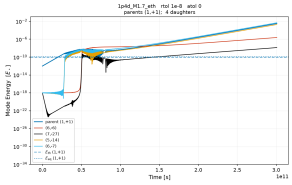
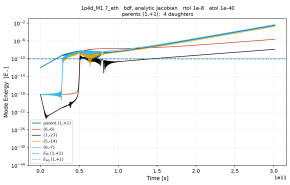
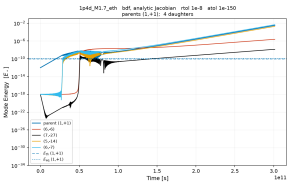
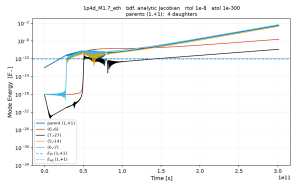
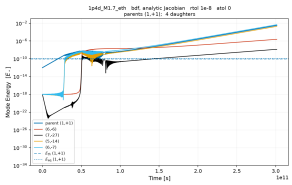
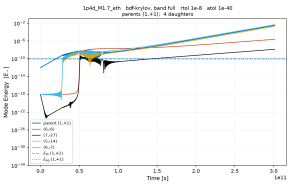
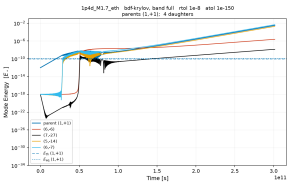
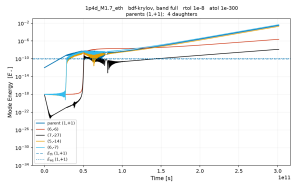
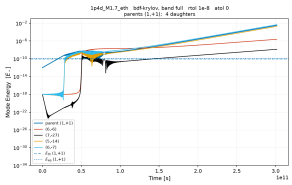
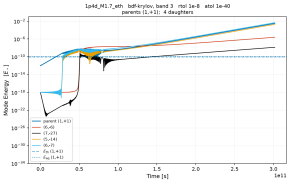
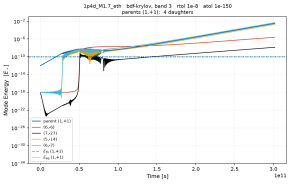
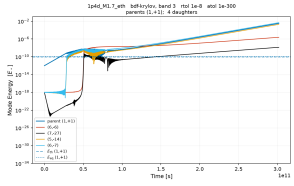
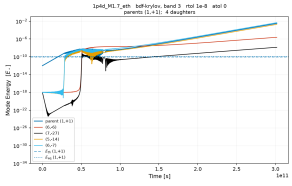
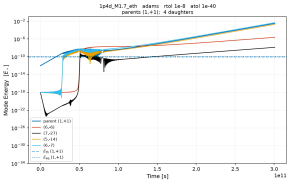
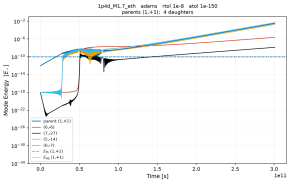
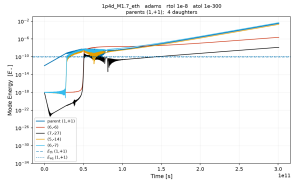
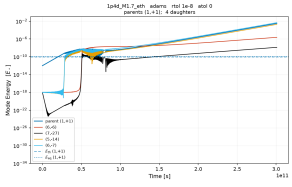
Model `dsct_M1.7_T7750`, rule eth, $N = 4$, cut 0.15, closure 0.005; 5 modes, 2 triplet(s).

Selecting pairs

(1,+1) → (6,−6) + (7,−27): $\Delta/\omega_a = -3.98 \times 10^{-5}$, $\kappa = +0.107$, $E_{\rm th} = 9.22 \times 10^{-11}$, $E_{\rm eq} = 1.23 \times 10^{-10}$

(1,+1) → (5,−14) + (6,−7): $\Delta/\omega_a = 1.51 \times 10^{-5}$, $\kappa = +0.369$, $E_{\rm th} = 1.57 \times 10^{-10}$, no $E_{\rm eq}$

Runaway near 3×10^{11} s.

	atol 10^{-40}	atol 10^{-150}	atol 10^{-300}	atol 0
rk8pd				
	7.58 M steps, 19.8 s, runaway	7.58 M steps, 17.7 s, runaway	7.58 M steps, 18.4 s, runaway	7.58 M steps, 19.8 s, runaway
BDF, dense analytic Jacobian				
	72.42 M steps, 74.9 s, runaway	72.41 M steps, 74.4 s, runaway	72.41 M steps, 122.1 s, runaway	72.41 M steps, 90.0 s, runaway
BDF-Krylov, band full				
	69.74 M steps, 133.2 s, runaway	69.63 M steps, 134.7 s, runaway	69.63 M steps, 109.1 s, runaway	69.63 M steps, 117.5 s, runaway
BDF-Krylov, band 3				
	66.98 M steps, 163.0 s, runaway	66.99 M steps, 172.6 s, runaway	66.99 M steps, 176.3 s, runaway	66.99 M steps, 170.4 s, runaway
Adams, fixed point				
	42.69 M steps, 64.2 s, runaway	42.69 M steps, 50.1 s, runaway	42.69 M steps, 52.2 s, runaway	42.69 M steps, 42.8 s, runaway

integrator	atol	stop	$t_{\rm stop}$ [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
rk8pd	10^{-40}	runaway	3.01×10^{11}	7.58 M	1.01 M	98.48 M	–	–	–	19.8
	10^{-150}	runaway	3.01×10^{11}	7.58 M	1.01 M	98.48 M	–	–	–	17.7
	10^{-300}	runaway	3.01×10^{11}	7.58 M	1.01 M	98.48 M	–	–	–	18.4
	0	runaway	3.01×10^{11}	7.58 M	1.01 M	98.48 M	–	–	–	19.8
BDF, dense analytic Jacobian	10^{-40}	runaway	3.01×10^{11}	72.42 M	911.7 k	78.90 M	1.27 M	78.90 M	–	74.9
	10^{-150}	runaway	3.01×10^{11}	72.41 M	918.0 k	78.93 M	1.27 M	78.93 M	–	74.4
	10^{-300}	runaway	3.01×10^{11}	72.41 M	918.0 k	78.93 M	1.27 M	78.93 M	–	122.1
	0	runaway	3.01×10^{11}	72.41 M	918.0 k	78.93 M	1.27 M	78.93 M	–	90.0
BDF-Krylov, band full	10^{-40}	runaway	3.02×10^{11}	69.74 M	989.6 k	163.10 M	1.19 M	76.22 M	75.01 M	133.2
	10^{-150}	runaway	3.02×10^{11}	69.63 M	1.00 M	162.99 M	1.19 M	76.17 M	74.96 M	134.7
	10^{-300}	runaway	3.02×10^{11}	69.63 M	1.00 M	162.99 M	1.19 M	76.17 M	74.96 M	109.1
	0	runaway	3.02×10^{11}	69.63 M	1.00 M	162.99 M	1.19 M	76.17 M	74.96 M	117.5
BDF-Krylov, band 3	10^{-40}	runaway	3.02×10^{11}	66.98 M	1.31 M	186.37 M	1.14 M	75.07 M	103.28 M	163.0
	10^{-150}	runaway	3.02×10^{11}	66.99 M	1.31 M	186.37 M	1.14 M	75.09 M	103.26 M	172.6
	10^{-300}	runaway	3.02×10^{11}	66.99 M	1.31 M	186.37 M	1.14 M	75.09 M	103.26 M	176.3
	0	runaway	3.02×10^{11}	66.99 M	1.31 M	186.37 M	1.14 M	75.09 M	103.26 M	170.4
Adams, fixed point	10^{-40}	runaway	3.01×10^{11}	42.69 M	2.54 M	68.34 M	–	68.34 M	–	64.2
	10^{-150}	runaway	3.01×10^{11}	42.69 M	2.54 M	68.34 M	–	68.34 M	–	50.1
	10^{-300}	runaway	3.01×10^{11}	42.69 M	2.54 M	68.34 M	–	68.34 M	–	52.2
	0	runaway	3.01×10^{11}	42.69 M	2.54 M	68.34 M	–	68.34 M	–	42.8

4.5 1p4d_M2.0T_eth

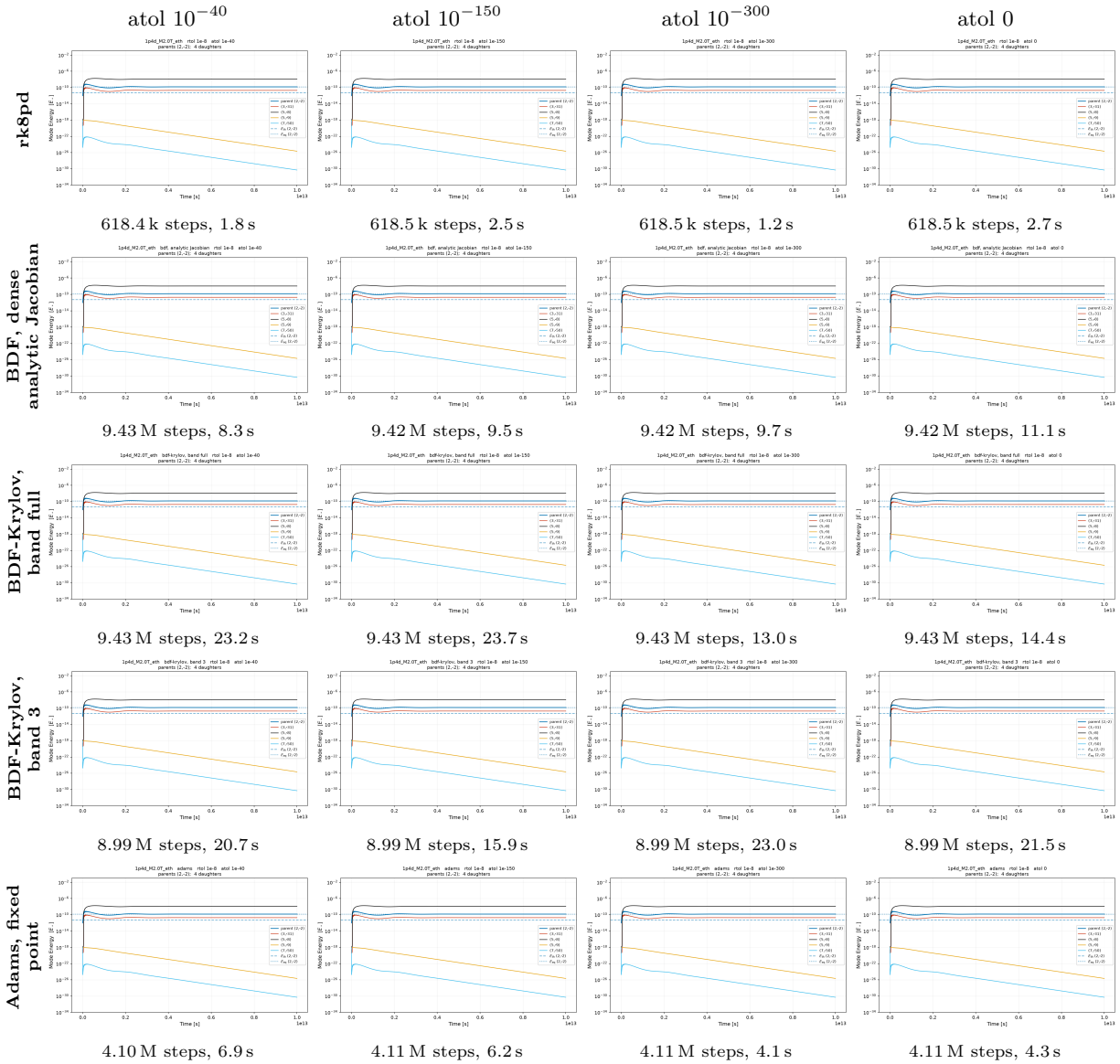
id	gen	(ℓ, n)	ω [rad s^{-1}]	γ [s^{-1}]
0	0	(2,−2)	7.959×10^{-4}	-6.697×10^{-11}
1	1	(3,−31)	1.287×10^{-4}	8.096×10^{-11}
2	1	(5,−8)	6.672×10^{-4}	7.367×10^{-13}
3	1	(5,−9)	6.216×10^{-4}	1.110×10^{-12}
4	1	(7,−50)	1.743×10^{-4}	2.546×10^{-9}

Model `dsct_M2.0_T7696`, rule eth, $N = 4$, cut 0.15, closure 0.005; 5 modes, 2 triplet(s).

Selecting pairs

(2,−2) → (3,−31) + (5,−8): $\Delta/\omega_a = -3.69 \times 10^{-6}$, $\kappa = +0.215$, $E_{\text{th}} = 4.87 \times 10^{-12}$, $E_{\text{eq}} = 1.50 \times 10^{-10}$

(2,−2) → (5,−9) + (7,−50): $\Delta/\omega_a = 7.51 \times 10^{-5}$, $\kappa = +0.053$, $E_{\text{th}} = 1.30 \times 10^{-9}$, $E_{\text{eq}} = 1.37 \times 10^{-9}$



integrator	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
rk8pd	10^{-40}	end	1.00×10^{13}	618.4 k	18.4 k	8.04 M	–	–	–	1.8
	10^{-150}	end	1.00×10^{13}	618.5 k	18.4 k	8.04 M	–	–	–	2.5
	10^{-300}	end	1.00×10^{13}	618.5 k	18.4 k	8.04 M	–	–	–	1.2
	0	end	1.00×10^{13}	618.5 k	18.4 k	8.04 M	–	–	–	2.7
BDF, dense analytic Jacobian	10^{-40}	end	1.00×10^{13}	9.43 M	89.6 k	10.13 M	167.6 k	10.13 M	–	8.3
	10^{-150}	end	1.00×10^{13}	9.42 M	89.5 k	10.12 M	167.4 k	10.12 M	–	9.5
	10^{-300}	end	1.00×10^{13}	9.42 M	89.5 k	10.12 M	167.4 k	10.12 M	–	9.7
	0	end	1.00×10^{13}	9.42 M	89.5 k	10.12 M	167.4 k	10.12 M	–	11.1
BDF-Krylov, band full	10^{-40}	end	1.00×10^{13}	9.43 M	89.3 k	21.30 M	167.6 k	10.12 M	9.49 M	23.2
	10^{-150}	end	1.00×10^{13}	9.43 M	89.6 k	21.29 M	167.5 k	10.12 M	9.49 M	23.7
	10^{-300}	end	1.00×10^{13}	9.43 M	89.6 k	21.29 M	167.5 k	10.12 M	9.49 M	13.0
	0	end	1.00×10^{13}	9.43 M	89.6 k	21.29 M	167.5 k	10.12 M	9.49 M	14.4
BDF-Krylov, band 3	10^{-40}	end	1.00×10^{13}	8.99 M	122.0 k	21.67 M	160.4 k	9.86 M	10.69 M	20.7
	10^{-150}	end	1.00×10^{13}	8.99 M	122.3 k	21.67 M	160.4 k	9.86 M	10.68 M	15.9
	10^{-300}	end	1.00×10^{13}	8.99 M	122.3 k	21.67 M	160.4 k	9.86 M	10.68 M	23.0
	0	end	1.00×10^{13}	8.99 M	122.3 k	21.67 M	160.4 k	9.86 M	10.68 M	21.5
Adams, fixed point	10^{-40}	end	1.00×10^{13}	4.10 M	287.6 k	6.43 M	–	6.42 M	–	6.9
	10^{-150}	end	1.00×10^{13}	4.11 M	283.9 k	6.44 M	–	6.44 M	–	6.2
	10^{-300}	end	1.00×10^{13}	4.11 M	283.9 k	6.44 M	–	6.44 M	–	4.1
	0	end	1.00×10^{13}	4.11 M	283.9 k	6.44 M	–	6.44 M	–	4.3

4.6 `1p4d_M2.0_eth`

id	gen	(ℓ, n)	ω [rad s^{-1}]	γ [s^{-1}]
0	0	(1,−1)	7.581×10^{-4}	-2.276×10^{-10}
1	1	(4,−38)	1.383×10^{-4}	2.224×10^{-10}
2	1	(5,−9)	6.198×10^{-4}	4.584×10^{-13}
3	1	(4,−8)	5.844×10^{-4}	4.902×10^{-13}
4	1	(5,−37)	1.736×10^{-4}	1.454×10^{-10}

Model `dsct_M2.0`, rule eth, $N = 4$, cut 0.15, closure 0.005; 5 modes, 2 triplet(s).

Selecting pairs

(1,−1) → (4,−38) + (5,−9): $\Delta/\omega_a = 6.73 \times 10^{-5}$, $\kappa = +0.342$, $E_{\mathrm{th}} = 1.33 \times 10^{-10}$, no E_{eq}

(1,−1) → (4,−8) + (5,−37): $\Delta/\omega_a = -1.23 \times 10^{-4}$, $\kappa = +0.171$, $E_{\mathrm{th}} = 2.46 \times 10^{-9}$, no E_{eq}

integrator	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
rk8pd	10^{-40}	end	1.00×10^{13}	2.41 M	469.8 k	31.29 M	–	–	–	5.2
	10^{-150}	end	1.00×10^{13}	2.41 M	469.8 k	31.29 M	–	–	–	10.5
	10^{-300}	end	1.00×10^{13}	2.41 M	469.8 k	31.29 M	–	–	–	7.6
	0	end	1.00×10^{13}	2.41 M	469.8 k	31.29 M	–	–	–	10.1
BDF, dense analytic Jacobian	10^{-40}	end	1.00×10^{13}	20.95 M	234.2 k	22.64 M	370.5 k	22.63 M	–	33.9
	10^{-150}	end	1.00×10^{13}	20.95 M	234.2 k	22.64 M	370.5 k	22.63 M	–	30.8
	10^{-300}	end	1.00×10^{13}	20.95 M	234.2 k	22.64 M	370.5 k	22.63 M	–	26.4
	0	end	1.00×10^{13}	20.95 M	234.2 k	22.64 M	370.5 k	22.63 M	–	34.6
BDF-Krylov, band full	10^{-40}	end	1.00×10^{13}	19.99 M	229.1 k	47.20 M	345.1 k	21.71 M	22.04 M	29.7
	10^{-150}	end	1.00×10^{13}	19.99 M	229.1 k	47.20 M	345.1 k	21.71 M	22.04 M	50.6
	10^{-300}	end	1.00×10^{13}	19.99 M	229.1 k	47.20 M	345.1 k	21.71 M	22.04 M	28.5
	0	end	1.00×10^{13}	19.99 M	229.1 k	47.20 M	345.1 k	21.71 M	22.04 M	49.4
BDF-Krylov, band 3	10^{-40}	end	1.00×10^{13}	19.90 M	240.9 k	47.57 M	343.7 k	21.69 M	23.47 M	30.2
	10^{-150}	end	1.00×10^{13}	19.89 M	240.0 k	47.56 M	343.5 k	21.68 M	23.48 M	50.7
	10^{-300}	end	1.00×10^{13}	19.89 M	240.0 k	47.56 M	343.5 k	21.68 M	23.48 M	35.2
	0	end	1.00×10^{13}	19.89 M	240.0 k	47.56 M	343.5 k	21.68 M	23.48 M	36.8
Adams, fixed point	10^{-40}	end	1.00×10^{13}	12.04 M	642.1 k	19.37 M	–	19.36 M	–	18.8
	10^{-150}	end	1.00×10^{13}	12.04 M	642.1 k	19.37 M	–	19.36 M	–	15.4
	10^{-300}	end	1.00×10^{13}	12.04 M	642.1 k	19.37 M	–	19.36 M	–	18.5
	0	end	1.00×10^{13}	12.04 M	642.1 k	19.37 M	–	19.36 M	–	11.4

4.7 2p_M1.7_eth

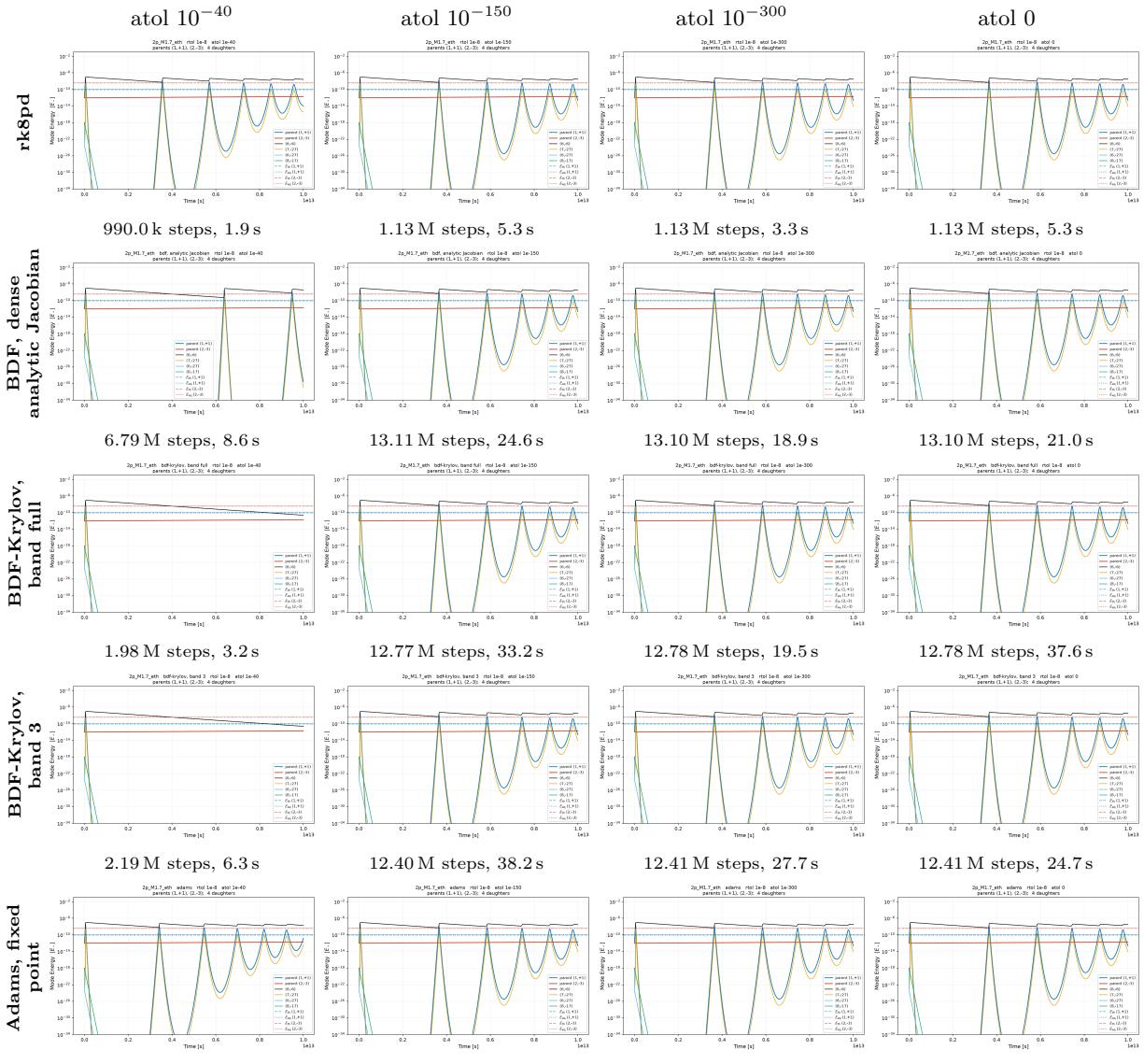
id	gen	(ℓ, n)	ω [rad s^{-1}]	γ [s^{-1}]
0	0	(1,+1)	1.402×10^{-3}	-1.150×10^{-10}
1	0	(2,−3)	8.582×10^{-4}	-3.458×10^{-14}
2	1	(6,−6)	1.070×10^{-3}	4.184×10^{-13}
3	1	(7,−27)	3.326×10^{-4}	8.612×10^{-10}
4	1	(6,−27)	2.895×10^{-4}	8.836×10^{-10}
5	1	(8,−17)	5.688×10^{-4}	3.001×10^{-11}

Model `dsct_M1.7_T7750`, rule eth, $N = 1$, cut 0.15, closure 0.005; 6 modes, 2 triplet(s).

Selecting pairs

(1,+1) → (6,−6) + (7,−27): $\Delta/\omega_a = -3.98 \times 10^{-5}$, $\kappa = +0.107$, $E_{\rm th} = 9.22 \times 10^{-11}$, $E_{\rm eq} = 1.23 \times 10^{-10}$

(2,−3) → (6,−27) + (8,−17): $\Delta/\omega_a = 5.08 \times 10^{-5}$, $\kappa = +0.156$, $E_{\rm th} = 3.75 \times 10^{-9}$, $E_{\rm eq} = 3.75 \times 10^{-9}$



integrator	atol	stop	$t_{\rm stop}$ [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
rk8pd	10^{-40}	end	1.00×10^{13}	990.0 k	104.5 k	12.87 M	–	–	–	1.9
	10^{-150}	end	1.00×10^{13}	1.13 M	128.5 k	14.69 M	–	–	–	5.3
	10^{-300}	end	1.00×10^{13}	1.13 M	128.6 k	14.69 M	–	–	–	3.3
	0	end	1.00×10^{13}	1.13 M	128.6 k	14.69 M	–	–	–	5.3
BDF, dense analytic Jacobian	10^{-40}	end	1.00×10^{13}	6.79 M	49.2 k	7.26 M	120.3 k	7.26 M	–	8.6
	10^{-150}	end	1.00×10^{13}	13.11 M	133.6 k	14.11 M	230.1 k	14.11 M	–	24.6
	10^{-300}	end	1.00×10^{13}	13.10 M	133.6 k	14.11 M	230.1 k	14.10 M	–	18.9
	0	end	1.00×10^{13}	13.10 M	133.6 k	14.11 M	230.1 k	14.10 M	–	21.0
BDF-Krylov, band full	10^{-40}	end	1.00×10^{13}	1.98 M	16.0 k	4.65 M	37.6 k	2.12 M	2.08 M	3.2
	10^{-150}	end	1.00×10^{13}	12.77 M	148.3 k	29.89 M	223.9 k	13.86 M	13.34 M	33.2
	10^{-300}	end	1.00×10^{13}	12.78 M	147.9 k	29.90 M	224.0 k	13.86 M	13.34 M	19.5
	0	end	1.00×10^{13}	12.78 M	147.9 k	29.90 M	224.0 k	13.86 M	13.34 M	37.6
BDF-Krylov, band 3	10^{-40}	end	1.00×10^{13}	2.19 M	24.0 k	5.58 M	41.3 k	2.39 M	2.90 M	6.3
	10^{-150}	end	1.00×10^{13}	12.40 M	183.1 k	31.21 M	217.9 k	13.67 M	16.01 M	38.2
	10^{-300}	end	1.00×10^{13}	12.41 M	183.7 k	31.21 M	217.9 k	13.67 M	16.01 M	27.7
	0	end	1.00×10^{13}	12.41 M	183.7 k	31.21 M	217.9 k	13.67 M	16.01 M	24.7
Adams, fixed point	10^{-40}	end	1.00×10^{13}	5.42 M	223.9 k	8.29 M	–	8.29 M	–	9.2
	10^{-150}	end	1.00×10^{13}	5.69 M	277.3 k	8.45 M	–	8.44 M	–	9.5
	10^{-300}	end	1.00×10^{13}	5.69 M	277.7 k	8.43 M	–	8.43 M	–	5.6
	0	end	1.00×10^{13}	5.69 M	277.7 k	8.43 M	–	8.43 M	–	6.3

4.8 2p_M2.0T_eth

id	gen	(ℓ, n)	ω [rad s^{-1}]	γ [s^{-1}]
0	0	(2,−3)	6.816×10^{-4}	-8.642×10^{-13}
1	0	(2,−2)	7.959×10^{-4}	-6.697×10^{-11}
2	1	(14,−48)	3.408×10^{-4}	1.376×10^{-9}
3	1	(3,−31)	1.287×10^{-4}	8.096×10^{-11}
4	1	(5,−8)	6.672×10^{-4}	7.367×10^{-13}

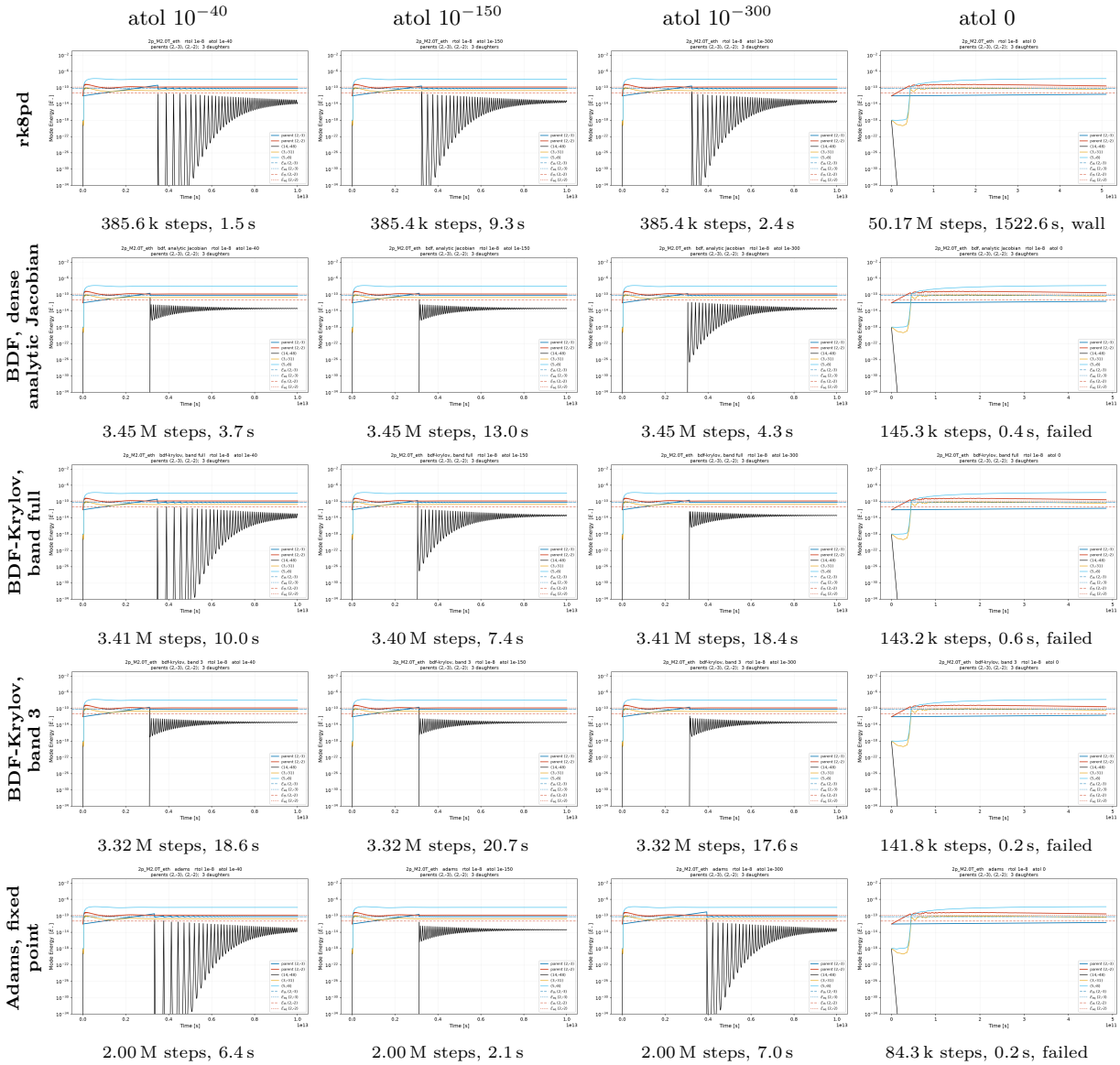
Model `dsct_M2.0_T7696`, rule eth, $N = 1$, cut 0.15, closure 0.005; 5 modes, 2 triplet(s).

Selecting pairs

(2,−3) → (14,−48) + (14,−48): $\Delta/\omega_a = -6.00 \times 10^{-6}$, $\kappa = -0.452$, $E_{\rm th} = 6.40 \times 10^{-11}$, $E_{\rm eq} = 6.40 \times 10^{-11}$

(2,−2) → (3,−31) + (5,−8): $\Delta/\omega_a = -3.69 \times 10^{-6}$, $\kappa = +0.215$, $E_{\rm th} = 4.87 \times 10^{-12}$, $E_{\rm eq} = 1.50 \times 10^{-10}$

Daughter (14,−48) decays to $E \sim 10^{-3600}$ in exact arithmetic; its regrowth time is set by each run's floor.



integrator	atol	stop	$t_{\rm stop}$ [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
rk8pd	10^{-40}	end	1.00×10^{13}	385.6 k	83.9 k	5.01 M	–	–	–	1.5
	10^{-150}	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	9.3
	10^{-300}	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	2.4
	0	wall	5.10×10^{11}	50.17 M	56.1 k	652.18 M	–	–	–	1522.6
BDF, dense analytic Jacobian	10^{-40}	end	1.00×10^{13}	3.45 M	57.6 k	3.84 M	72.5 k	3.83 M	–	3.7
	10^{-150}	end	1.00×10^{13}	3.45 M	57.4 k	3.84 M	72.5 k	3.83 M	–	13.0
	10^{-300}	end	1.00×10^{13}	3.45 M	57.3 k	3.84 M	72.5 k	3.83 M	–	4.3
	0	failed	4.85×10^{11}	145.3 k	3.9 k	167.3 k	3.2 k	167.1 k	–	0.4
BDF-Krylov, band full	10^{-40}	end	1.00×10^{13}	3.41 M	60.6 k	7.96 M	71.0 k	3.77 M	3.48 M	10.0
	10^{-150}	end	1.00×10^{13}	3.40 M	61.2 k	7.95 M	71.0 k	3.76 M	3.48 M	7.4
	10^{-300}	end	1.00×10^{13}	3.41 M	60.7 k	7.96 M	71.0 k	3.77 M	3.48 M	18.4
	0	failed	4.85×10^{11}	143.2 k	3.7 k	342.1 k	3.1 k	162.9 k	148.4 k	0.6
BDF-Krylov, band 3	10^{-40}	end	1.00×10^{13}	3.32 M	67.5 k	8.56 M	69.7 k	3.72 M	4.35 M	18.6
	10^{-150}	end	1.00×10^{13}	3.32 M	68.1 k	8.57 M	69.6 k	3.71 M	4.36 M	20.7
	10^{-300}	end	1.00×10^{13}	3.32 M	68.0 k	8.57 M	69.7 k	3.72 M	4.37 M	17.6
	0	failed	4.85×10^{11}	141.8 k	4.0 k	367.9 k	3.0 k	162.9 k	183.5 k	0.2
Adams, fixed point	10^{-40}	end	1.00×10^{13}	2.00 M	131.7 k	3.30 M	–	3.30 M	–	6.4
	10^{-150}	end	1.00×10^{13}	2.00 M	132.1 k	3.30 M	–	3.29 M	–	2.1
	10^{-300}	end	1.00×10^{13}	2.00 M	132.3 k	3.30 M	–	3.29 M	–	7.0
	0	failed	4.85×10^{11}	84.3 k	5.9 k	138.5 k	–	138.3 k	–	0.2

4.9 2p_M2.0_eeq

id	gen	(ℓ, n)	ω [rad s^{-1}]	γ [s^{-1}]
0	0	(1,−1)	7.581×10^{-4}	-2.276×10^{-10}
1	0	(2,−2)	7.710×10^{-4}	-1.619×10^{-10}
2	1	(5,−9)	6.198×10^{-4}	4.584×10^{-13}
3	1	(6,−56)	1.377×10^{-4}	2.817×10^{-9}
4	1	(6,−11)	5.949×10^{-4}	1.643×10^{-12}
5	1	(8,−57)	1.762×10^{-4}	4.129×10^{-9}

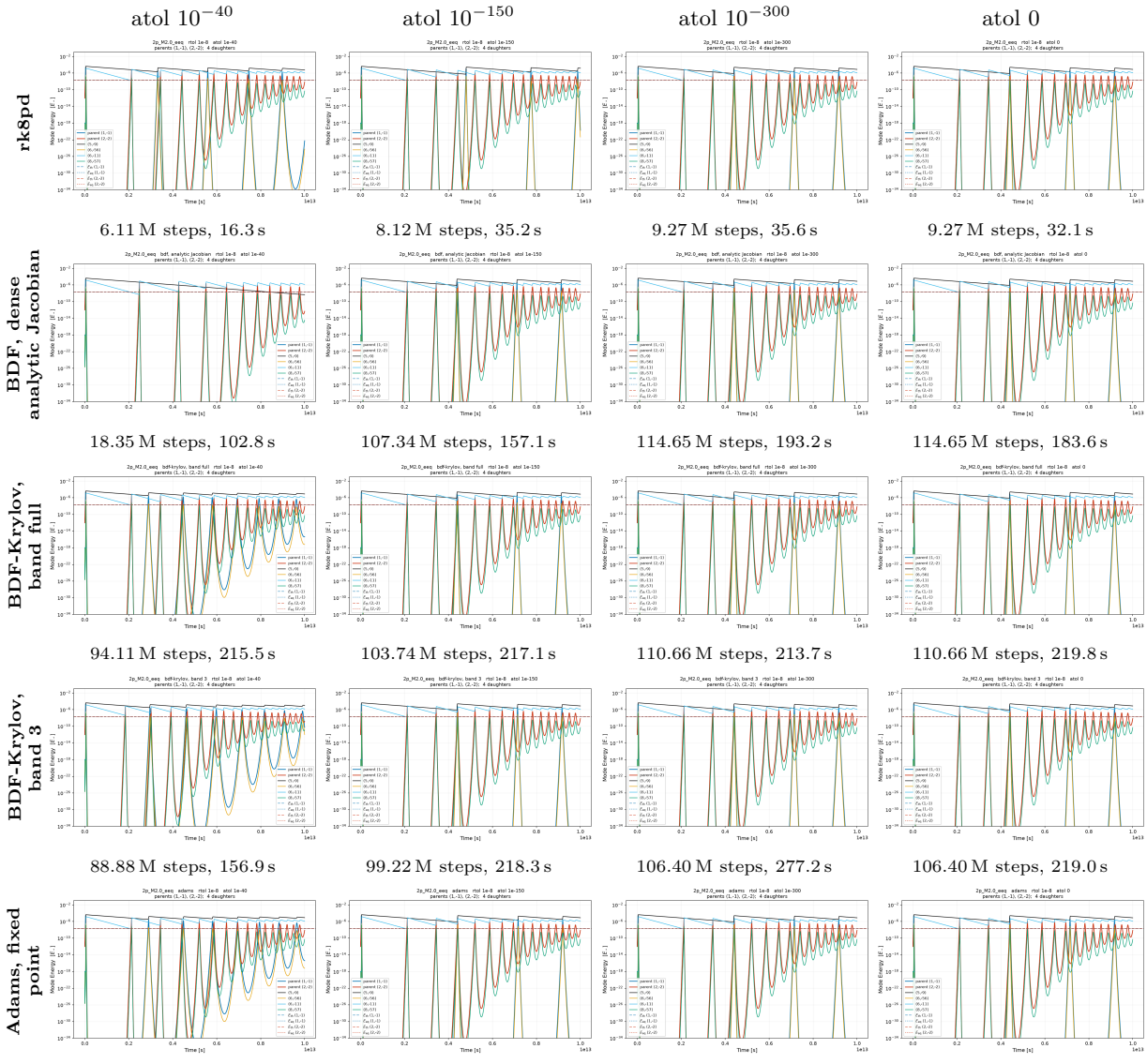
Model `dsct_M2.0`, rule eeq, $N = 1$, cut 0.15, closure 0.005; 6 modes, 2 triplet(s).

Selecting pairs

(1,−1) → (5,−9) + (6,−56): $\Delta/\omega_a = -6.83 \times 10^{-4}$, $\kappa = +0.085$, $E_{\rm th} = 1.75 \times 10^{-8}$, $E_{\rm eq} = 2.07 \times 10^{-8}$

(2,−2) → (6,−11) + (8,−57): $\Delta/\omega_a = 1.36 \times 10^{-4}$, $\kappa = +0.023$, $E_{\rm th} = 1.97 \times 10^{-8}$, $E_{\rm eq} = 2.13 \times 10^{-8}$

Deep dips between bursts.



6.11 M steps, 16.3 s | 8.12 M steps, 35.2 s | 9.27 M steps, 35.6 s | 9.27 M steps, 32.1 s

18.35 M steps, 102.8 s | 107.34 M steps, 157.1 s | 114.65 M steps, 193.2 s | 114.65 M steps, 183.6 s

94.11 M steps, 215.5 s | 103.74 M steps, 217.1 s | 110.66 M steps, 213.7 s | 110.66 M steps, 219.8 s

88.88 M steps, 156.9 s | 99.22 M steps, 218.3 s | 106.40 M steps, 277.2 s | 106.40 M steps, 219.0 s

40.39 M steps, 52.7 s | 43.82 M steps, 57.0 s | 46.13 M steps, 49.8 s | 46.13 M steps, 48.4 s

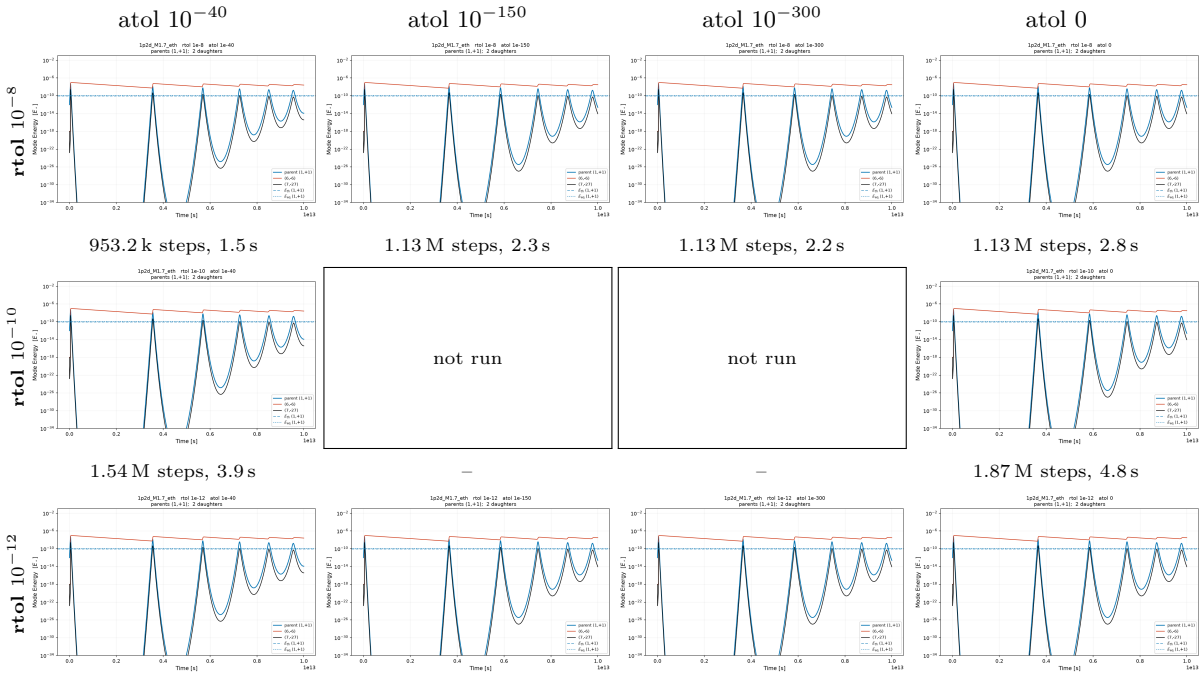
integrator	atol	stop	$t_{\rm stop}$ [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
rk8pd	10^{-40}	end	1.00×10^{13}	6.11 M	535.4 k	79.40 M	–	–	–	16.3
	10^{-150}	end	1.00×10^{13}	8.12 M	944.9 k	105.61 M	–	–	–	35.2
	10^{-300}	end	1.00×10^{13}	9.27 M	1.04 M	120.46 M	–	–	–	35.6
	0	end	1.00×10^{13}	9.27 M	1.04 M	120.46 M	–	–	–	32.1
BDF, dense analytic Jacobian	10^{-40}	end	1.00×10^{13}	18.35 M	206.2 k	20.00 M	382.9 k	20.00 M	–	102.8
	10^{-150}	end	1.00×10^{13}	107.34 M	674.9 k	113.72 M	1.81 M	113.72 M	–	157.1
	10^{-300}	end	1.00×10^{13}	114.65 M	735.3 k	121.52 M	1.93 M	121.52 M	–	193.2
	0	end	1.00×10^{13}	114.65 M	735.3 k	121.52 M	1.93 M	121.52 M	–	183.6
BDF-Krylov, band full	10^{-40}	end	1.00×10^{13}	94.11 M	669.5 k	220.88 M	1.58 M	100.68 M	101.20 M	215.5
	10^{-150}	end	1.00×10^{13}	103.74 M	765.1 k	239.23 M	1.74 M	110.95 M	107.36 M	217.1
	10^{-300}	end	1.00×10^{13}	110.66 M	843.3 k	254.79 M	1.86 M	118.47 M	114.00 M	213.7
	0	end	1.00×10^{13}	110.66 M	843.3 k	254.79 M	1.86 M	118.47 M	114.00 M	219.8
BDF-Krylov, band 3	10^{-40}	end	1.00×10^{13}	88.88 M	917.3 k	219.52 M	1.50 M	96.67 M	112.37 M	156.9
	10^{-150}	end	1.00×10^{13}	99.22 M	1.06 M	241.99 M	1.67 M	107.98 M	122.31 M	218.3
	10^{-300}	end	1.00×10^{13}	106.40 M	1.15 M	258.82 M	1.79 M	115.88 M	130.40 M	277.2
	0	end	1.00×10^{13}	106.40 M	1.15 M	258.82 M	1.79 M	115.88 M	130.40 M	219.0
Adams, fixed point	10^{-40}	end	1.00×10^{13}	40.39 M	1.44 M	58.85 M	–	58.84 M	–	52.7
	10^{-150}	end	1.00×10^{13}	43.82 M	1.91 M	64.54 M	–	64.54 M	–	57.0
	10^{-300}	end	1.00×10^{13}	46.13 M	2.08 M	67.26 M	–	67.26 M	–	49.8
	0	end	1.00×10^{13}	46.13 M	2.08 M	67.26 M	–	67.26 M	–	48.4

A rk8pd: relative tolerance

These pages use the same layout as the network pages, but the rows are rtol instead of integrators. rtol = 10^{-10} was run only at atol = 0 and 10^{-40}.

A.1 `1p2d_M1.7_eth`: rk8pd, rtol × atol

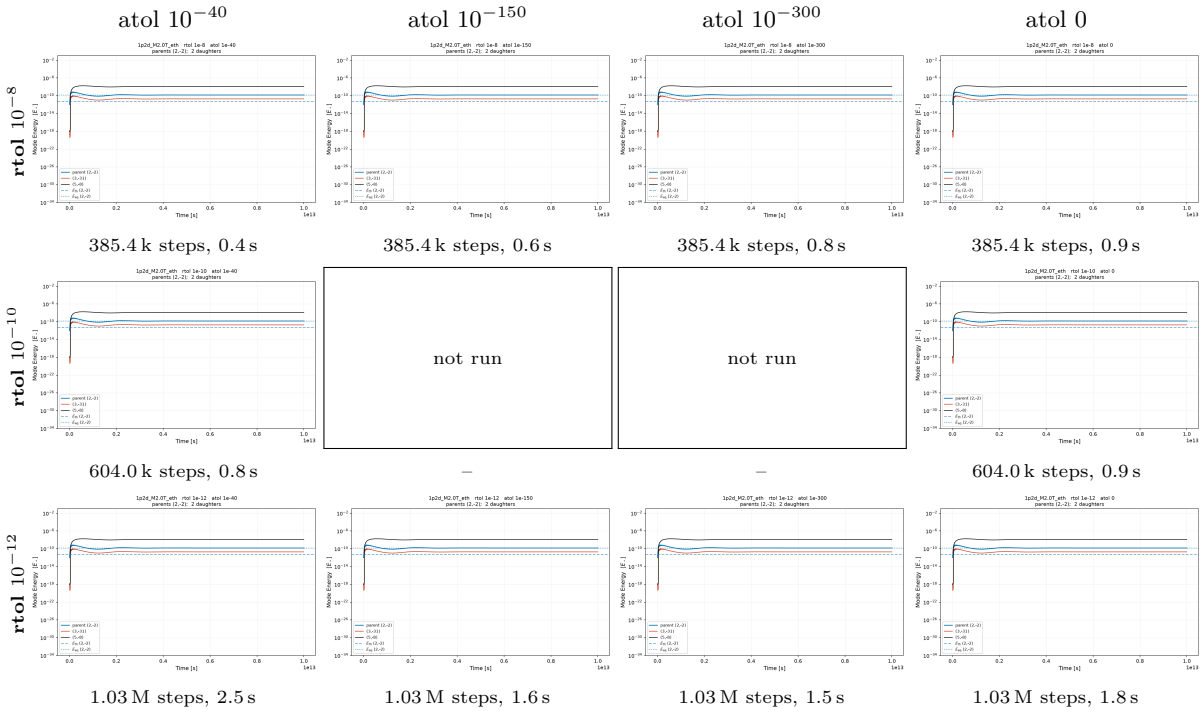
See §?? for the network. The second burst disagreed between rk8pd and BDF in the earlier sweep.



rtol	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
10^{-8}	10^{-40}	end	1.00×10^{13}	953.2 k	101.3 k	12.39 M	–	–	–	1.5
	10^{-150}	end	1.00×10^{13}	1.13 M	126.8 k	14.65 M	–	–	–	2.3
	10^{-300}	end	1.00×10^{13}	1.13 M	126.8 k	14.65 M	–	–	–	2.2
	0	end	1.00×10^{13}	1.13 M	126.8 k	14.65 M	–	–	–	2.8
10^{-10}	10^{-40}	end	1.00×10^{13}	1.54 M	110.1 k	20.04 M	–	–	–	3.9
	10^{-150}	not run								
	10^{-300}	not run								
	0	end	1.00×10^{13}	1.87 M	140.6 k	24.25 M	–	–	–	4.8
10^{-12}	10^{-40}	end	1.00×10^{13}	2.56 M	116.1 k	33.27 M	–	–	–	4.4
	10^{-150}	end	1.00×10^{13}	3.16 M	153.8 k	41.12 M	–	–	–	4.5
	10^{-300}	end	1.00×10^{13}	3.16 M	153.8 k	41.12 M	–	–	–	4.6
	0	end	1.00×10^{13}	3.16 M	153.8 k	41.12 M	–	–	–	7.8

A.2 `1p2d_M2.0T_eth`: rk8pd, rtol × atol

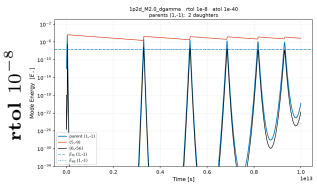
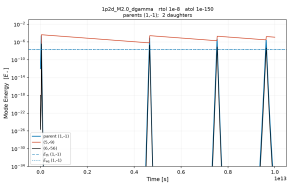
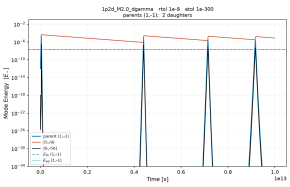
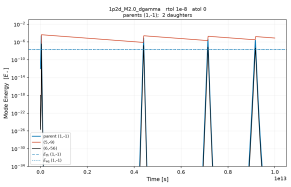
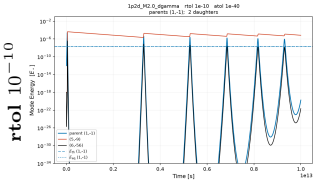
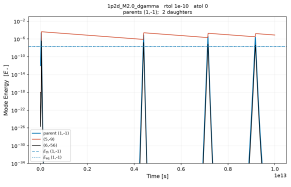
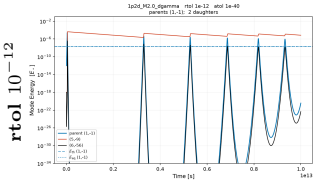
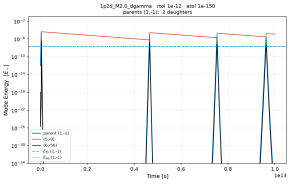
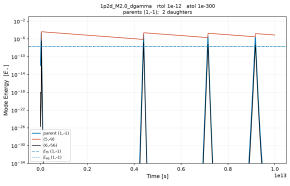
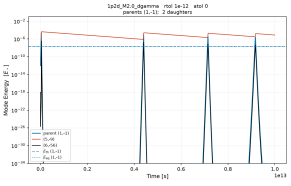
See §?? for the network.



rtol	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
10^{-8}	10^{-40}	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	0.4
	10^{-150}	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	0.6
	10^{-300}	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	0.8
	0	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	0.9
10^{-10}	10^{-40}	end	1.00×10^{13}	604.0 k	87.8 k	7.85 M	–	–	–	0.8
	10^{-150}	not run								
	10^{-300}	not run								
	0	end	1.00×10^{13}	604.0 k	87.8 k	7.85 M	–	–	–	0.9
10^{-12}	10^{-40}	end	1.00×10^{13}	1.03 M	131.8 k	13.36 M	–	–	–	2.5
	10^{-150}	end	1.00×10^{13}	1.03 M	131.8 k	13.36 M	–	–	–	1.6
	10^{-300}	end	1.00×10^{13}	1.03 M	131.8 k	13.36 M	–	–	–	1.5
	0	end	1.00×10^{13}	1.03 M	131.8 k	13.36 M	–	–	–	1.8

A.3 `1p2d_M2.0_dgamma`: rk8pd, rtol × atol

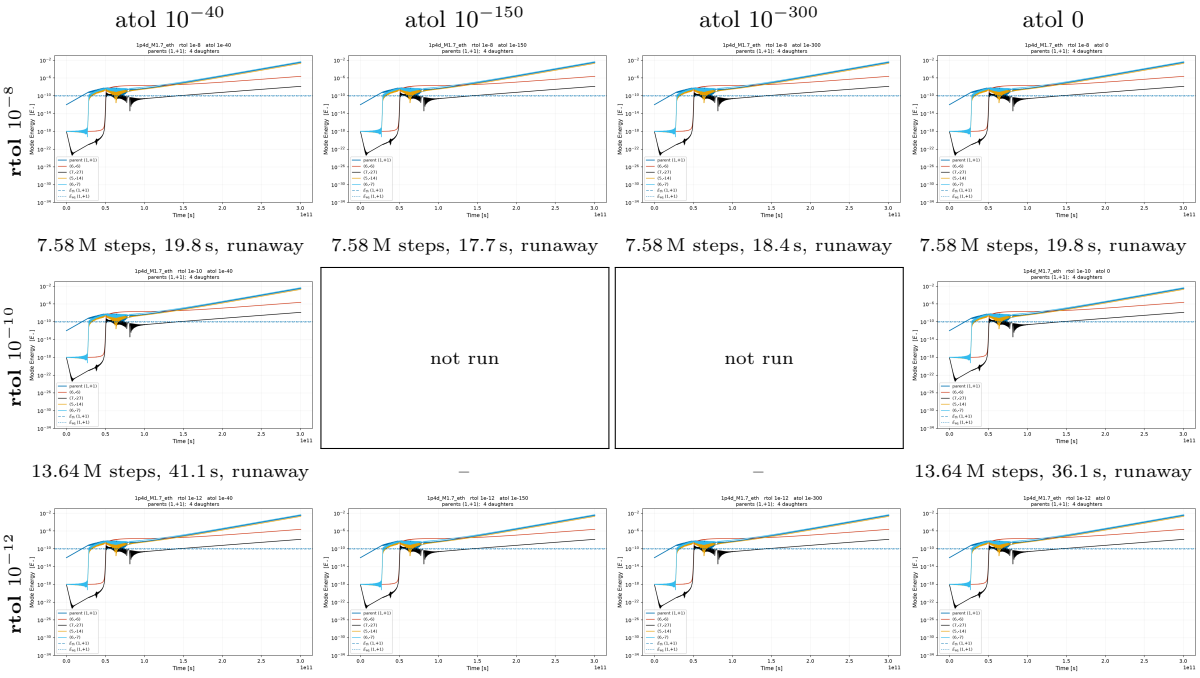
See §?? for the network. Deep dips between bursts.

	atol 10^{-40}	atol 10^{-150}	atol 10^{-300}	atol 0
rtol 10^{-8}				
	6.61 M steps, 9.5 s	8.24 M steps, 15.6 s	9.26 M steps, 13.5 s	9.26 M steps, 20.4 s
rtol 10^{-10}		not run	not run	
	10.60 M steps, 19.0 s	–	–	15.47 M steps, 35.0 s
rtol 10^{-12}				
	17.32 M steps, 34.3 s	22.42 M steps, 33.3 s	26.08 M steps, 32.1 s	26.08 M steps, 50.2 s

rtol	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
10^{-8}	10^{-40}	end	1.00×10^{13}	6.61 M	575.7 k	85.89 M	–	–	–	9.5
	10^{-150}	end	1.00×10^{13}	8.24 M	937.9 k	107.11 M	–	–	–	15.6
	10^{-300}	end	1.00×10^{13}	9.26 M	1.04 M	120.43 M	–	–	–	13.5
	0	end	1.00×10^{13}	9.26 M	1.04 M	120.43 M	–	–	–	20.4
10^{-10}	10^{-40}	end	1.00×10^{13}	10.60 M	653.6 k	137.86 M	–	–	–	19.0
	10^{-150}	not run								
	10^{-300}	not run								
	0	end	1.00×10^{13}	15.47 M	1.23 M	201.05 M	–	–	–	35.0
10^{-12}	10^{-40}	end	1.00×10^{13}	17.32 M	702.1 k	225.19 M	–	–	–	34.3
	10^{-150}	end	1.00×10^{13}	22.42 M	1.16 M	291.47 M	–	–	–	33.3
	10^{-300}	end	1.00×10^{13}	26.08 M	1.34 M	339.08 M	–	–	–	32.1
	0	end	1.00×10^{13}	26.08 M	1.34 M	339.08 M	–	–	–	50.2

A.4 `1p4d_M1.7_eth`: rk8pd, rtol × atol

See §?? for the network. Runaway near 3×10^{11} s.



7.58 M steps, 19.8 s, runaway | 7.58 M steps, 17.7 s, runaway | 7.58 M steps, 18.4 s, runaway | 7.58 M steps, 19.8 s, runaway

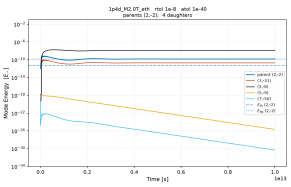
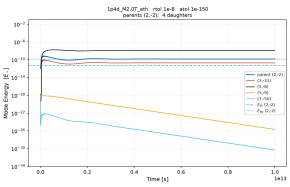
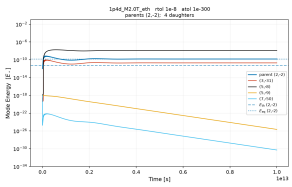
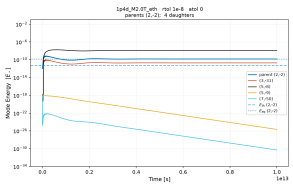
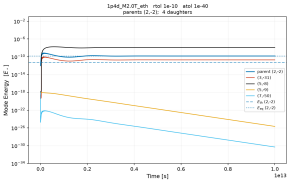
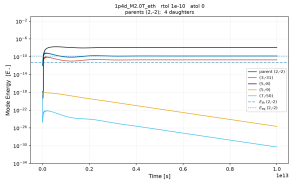
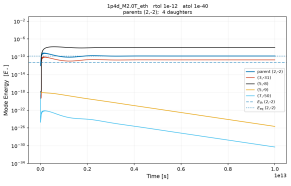
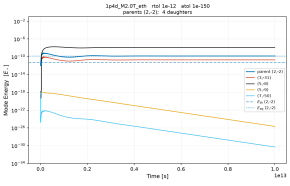
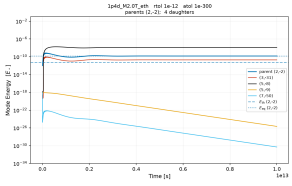
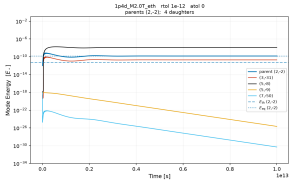
13.64 M steps, 41.1 s, runaway | – | – | 13.64 M steps, 36.1 s, runaway

22.63 M steps, 58.6 s, runaway | 22.63 M steps, 57.4 s, runaway | 22.63 M steps, 57.3 s, runaway | 22.63 M steps, 82.7 s, runaway

rtol	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
10^{-8}	10^{-40}	runaway	3.01×10^{11}	7.58 M	1.01 M	98.48 M	–	–	–	19.8
	10^{-150}	runaway	3.01×10^{11}	7.58 M	1.01 M	98.48 M	–	–	–	17.7
	10^{-300}	runaway	3.01×10^{11}	7.58 M	1.01 M	98.48 M	–	–	–	18.4
	0	runaway	3.01×10^{11}	7.58 M	1.01 M	98.48 M	–	–	–	19.8
10^{-10}	10^{-40}	runaway	3.01×10^{11}	13.64 M	1.93 M	177.26 M	–	–	–	41.1
	10^{-150}	not run								
	10^{-300}	not run								
	0	runaway	3.01×10^{11}	13.64 M	1.93 M	177.29 M	–	–	–	36.1
10^{-12}	10^{-40}	runaway	3.01×10^{11}	22.63 M	2.32 M	294.20 M	–	–	–	58.6
	10^{-150}	runaway	3.01×10^{11}	22.63 M	2.33 M	294.20 M	–	–	–	57.4
	10^{-300}	runaway	3.01×10^{11}	22.63 M	2.33 M	294.20 M	–	–	–	57.3
	0	runaway	3.01×10^{11}	22.63 M	2.33 M	294.20 M	–	–	–	82.7

A.5 `1p4d_M2.0T_eth`: rk8pd, rtol × atol

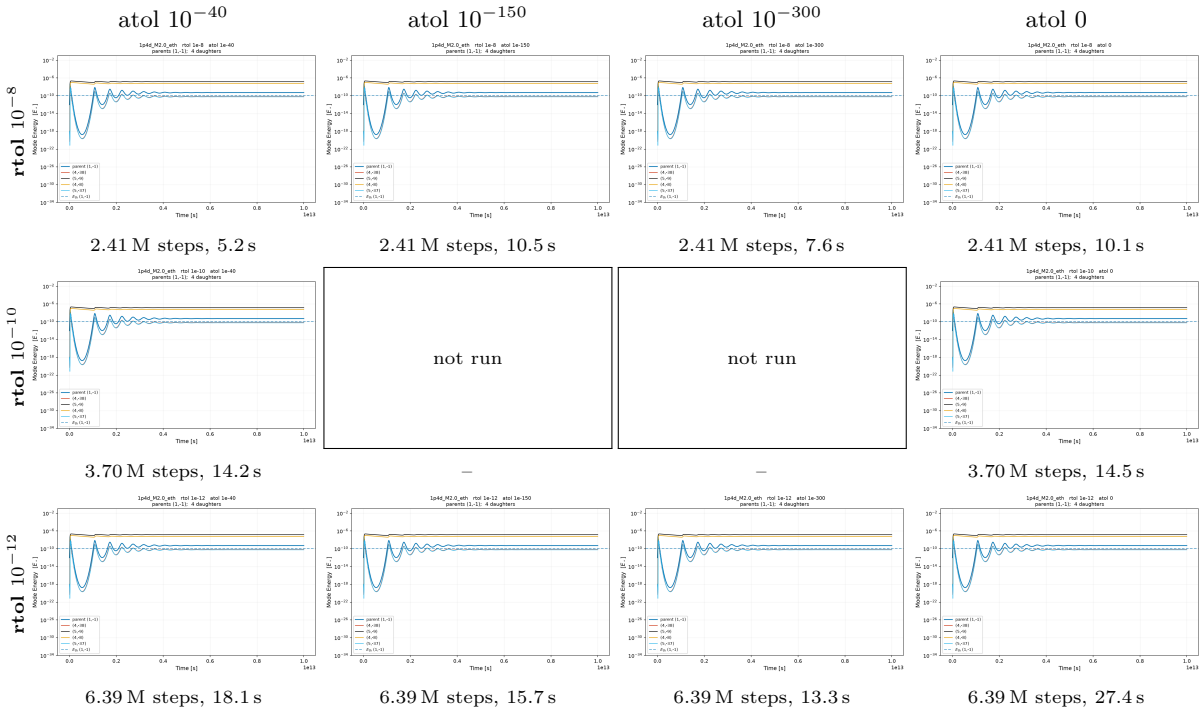
See §?? for the network.

	atol 10^{-40}	atol 10^{-150}	atol 10^{-300}	atol 0
rtol 10^{-8}				
	618.4 k steps, 1.8 s	618.5 k steps, 2.5 s	618.5 k steps, 1.2 s	618.5 k steps, 2.7 s
rtol 10^{-10}		not run	not run	
	1.47 M steps, 4.5 s	–	–	1.47 M steps, 4.6 s
rtol 10^{-12}				
	2.37 M steps, 9.4 s	2.37 M steps, 6.6 s	2.37 M steps, 4.8 s	2.37 M steps, 9.2 s

rtol	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
10^{-8}	10^{-40}	end	1.00×10^{13}	618.4 k	18.4 k	8.04 M	–	–	–	1.8
	10^{-150}	end	1.00×10^{13}	618.5 k	18.4 k	8.04 M	–	–	–	2.5
	10^{-300}	end	1.00×10^{13}	618.5 k	18.4 k	8.04 M	–	–	–	1.2
	0	end	1.00×10^{13}	618.5 k	18.4 k	8.04 M	–	–	–	2.7
10^{-10}	10^{-40}	end	1.00×10^{13}	1.47 M	287.4 k	19.11 M	–	–	–	4.5
	10^{-150}	not run								
	10^{-300}	not run								
	0	end	1.00×10^{13}	1.47 M	287.3 k	19.11 M	–	–	–	4.6
10^{-12}	10^{-40}	end	1.00×10^{13}	2.37 M	303.5 k	30.87 M	–	–	–	9.4
	10^{-150}	end	1.00×10^{13}	2.37 M	303.3 k	30.87 M	–	–	–	6.6
	10^{-300}	end	1.00×10^{13}	2.37 M	303.3 k	30.87 M	–	–	–	4.8
	0	end	1.00×10^{13}	2.37 M	303.3 k	30.87 M	–	–	–	9.2

A.6 `1p4d_M2.0_eth`: rk8pd, rtol × atol

See §?? for the network.



2.41 M steps, 5.2 s | 2.41 M steps, 10.5 s | 2.41 M steps, 7.6 s | 2.41 M steps, 10.1 s

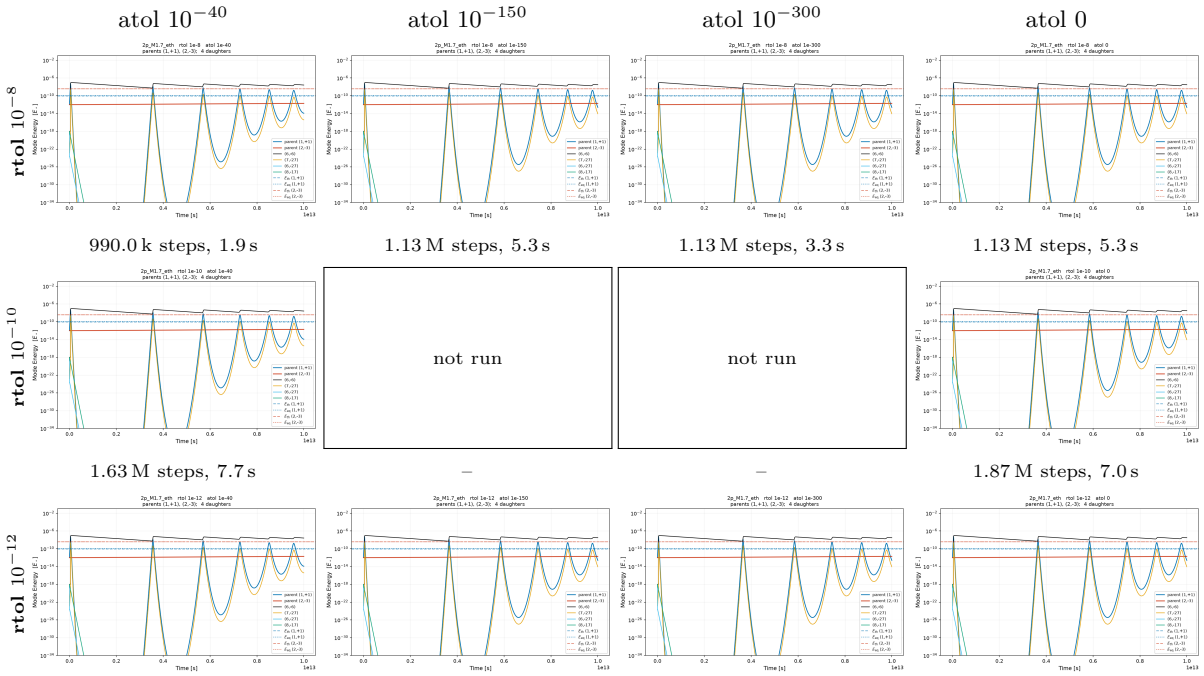
3.70 M steps, 14.2 s | – | – | 3.70 M steps, 14.5 s

6.39 M steps, 18.1 s | 6.39 M steps, 15.7 s | 6.39 M steps, 13.3 s | 6.39 M steps, 27.4 s

rtol	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
10^{-8}	10^{-40}	end	1.00×10^{13}	2.41 M	469.8 k	31.29 M	–	–	–	5.2
	10^{-150}	end	1.00×10^{13}	2.41 M	469.8 k	31.29 M	–	–	–	10.5
	10^{-300}	end	1.00×10^{13}	2.41 M	469.8 k	31.29 M	–	–	–	7.6
	0	end	1.00×10^{13}	2.41 M	469.8 k	31.29 M	–	–	–	10.1
10^{-10}	10^{-40}	end	1.00×10^{13}	3.70 M	439.1 k	48.05 M	–	–	–	14.2
	10^{-150}	not run								
	10^{-300}	not run								
	0	end	1.00×10^{13}	3.70 M	439.1 k	48.05 M	–	–	–	14.5
10^{-12}	10^{-40}	end	1.00×10^{13}	6.39 M	647.9 k	83.13 M	–	–	–	18.1
	10^{-150}	end	1.00×10^{13}	6.39 M	647.7 k	83.13 M	–	–	–	15.7
	10^{-300}	end	1.00×10^{13}	6.39 M	647.7 k	83.13 M	–	–	–	13.3
	0	end	1.00×10^{13}	6.39 M	647.7 k	83.13 M	–	–	–	27.4

A.7 `2p_M1.7_eth`: rk8pd, rtol × atol

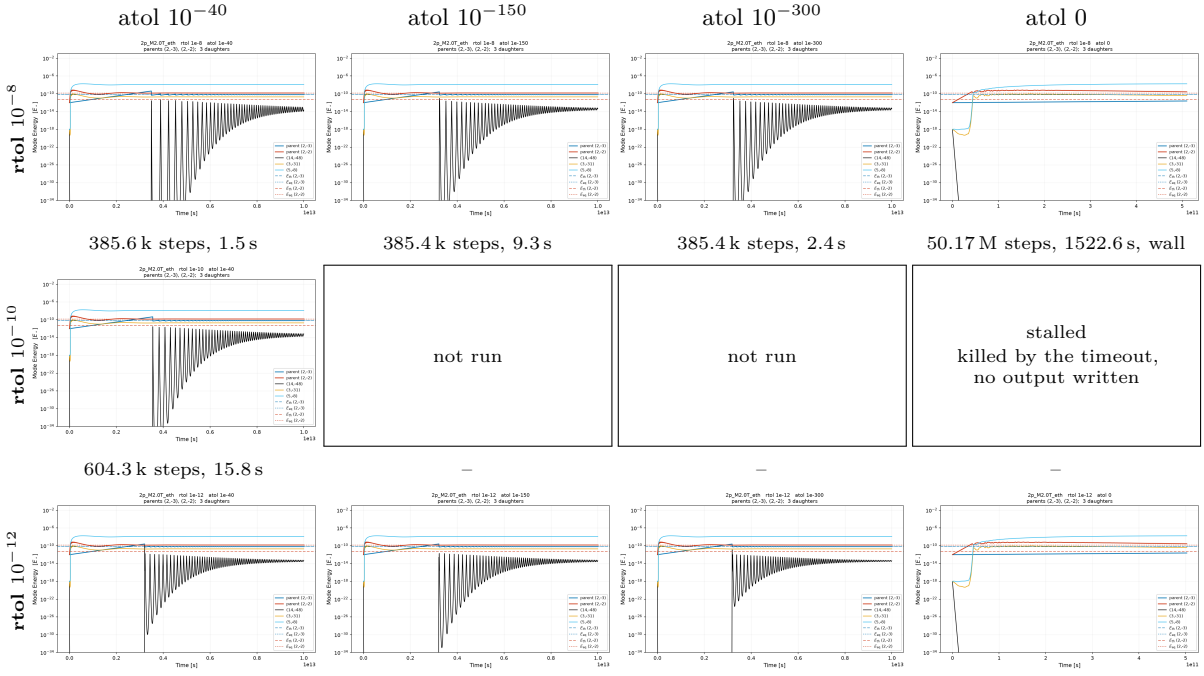
See §?? for the network.



rtol	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
10^{-8}	10^{-40}	end	1.00×10^{13}	990.0 k	104.5 k	12.87 M	–	–	–	1.9
	10^{-150}	end	1.00×10^{13}	1.13 M	128.5 k	14.69 M	–	–	–	5.3
	10^{-300}	end	1.00×10^{13}	1.13 M	128.6 k	14.69 M	–	–	–	3.3
	0	end	1.00×10^{13}	1.13 M	128.6 k	14.69 M	–	–	–	5.3
10^{-10}	10^{-40}	end	1.00×10^{13}	1.63 M	125.6 k	21.14 M	–	–	–	7.7
	10^{-150}	not run								
	10^{-300}	not run								
	0	end	1.00×10^{13}	1.87 M	143.7 k	24.31 M	–	–	–	7.0
10^{-12}	10^{-40}	end	1.00×10^{13}	2.68 M	128.7 k	34.85 M	–	–	–	11.8
	10^{-150}	end	1.00×10^{13}	3.17 M	158.2 k	41.21 M	–	–	–	9.0
	10^{-300}	end	1.00×10^{13}	3.17 M	158.5 k	41.21 M	–	–	–	7.0
	0	end	1.00×10^{13}	3.17 M	158.5 k	41.21 M	–	–	–	12.2

A.8 `2p_M2.0T_eth`: rk8pd, rtol × atol

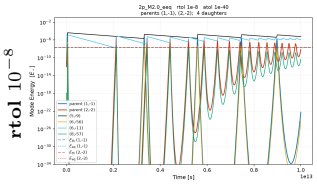
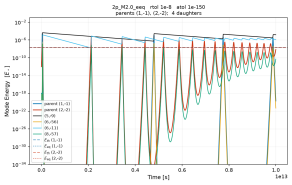
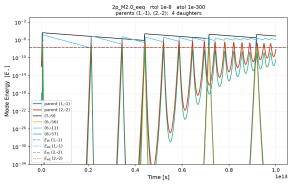
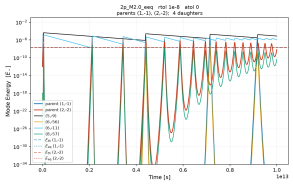
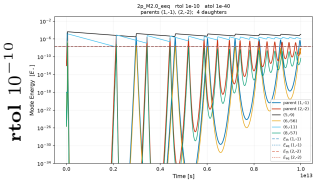
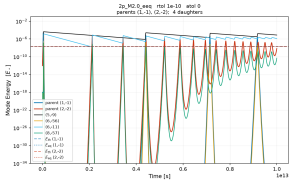
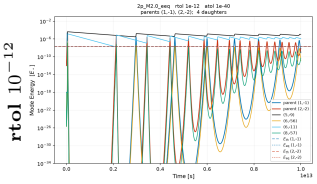
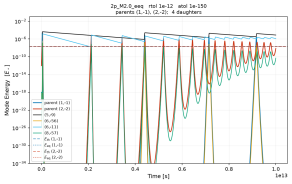
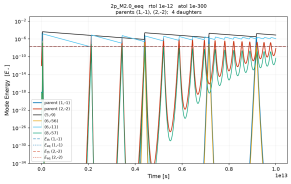
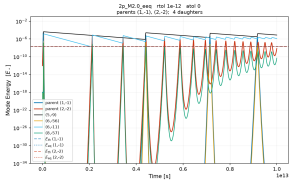
See §?? for the network. Daughter (14,−48) decays to $E \sim 10^{-3600}$ in exact arithmetic; its regrowth time is set by each run's floor.



rtol	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
10^{-8}	10^{-40}	end	1.00×10^{13}	385.6 k	83.9 k	5.01 M	–	–	–	1.5
	10^{-150}	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	9.3
	10^{-300}	end	1.00×10^{13}	385.4 k	83.7 k	5.01 M	–	–	–	2.4
	0	wall	5.10×10^{11}	50.17 M	56.1 k	652.18 M	–	–	–	1522.6
10^{-10}	10^{-40}	end	1.00×10^{13}	604.3 k	87.9 k	7.86 M	–	–	–	15.8
	10^{-150}	not run								
	10^{-300}	not run								
	0	stalled	no output written							
10^{-12}	10^{-40}	end	1.00×10^{13}	1.03 M	132.0 k	13.36 M	–	–	–	11.0
	10^{-150}	end	1.00×10^{13}	1.03 M	132.1 k	13.37 M	–	–	–	14.6
	10^{-300}	end	1.00×10^{13}	1.03 M	132.2 k	13.37 M	–	–	–	10.5
	0	wall	5.03×10^{11}	24.77 M	5.09 M	321.99 M	–	–	–	1715.4

A.9 `2p_M2.0_eeq`: rk8pd, rtol × atol

See §?? for the network. Deep dips between bursts.

	atol 10^{-40}	atol 10^{-150}	atol 10^{-300}	atol 0
rtol 10^{-8}				
	6.11 M steps, 16.3 s	8.12 M steps, 35.2 s	9.27 M steps, 35.6 s	9.27 M steps, 32.1 s
rtol 10^{-10}		not run	not run	
	12.20 M steps, 33.3 s	–	–	15.47 M steps, 52.3 s
rtol 10^{-12}				
	20.56 M steps, 62.8 s	24.05 M steps, 71.5 s	26.09 M steps, 70.3 s	26.09 M steps, 82.0 s

rtol	atol	stop	t_{stop} [s]	steps	rejected	RHS	Jac.	NL it.	lin. it.	wall [s]
10^{-8}	10^{-40}	end	1.00×10^{13}	6.11 M	535.4 k	79.40 M	–	–	–	16.3
	10^{-150}	end	1.00×10^{13}	8.12 M	944.9 k	105.61 M	–	–	–	35.2
	10^{-300}	end	1.00×10^{13}	9.27 M	1.04 M	120.46 M	–	–	–	35.6
	0	end	1.00×10^{13}	9.27 M	1.04 M	120.46 M	–	–	–	32.1
10^{-10}	10^{-40}	end	1.00×10^{13}	12.20 M	774.2 k	158.63 M	–	–	–	33.3
	10^{-150}	not run								
	10^{-300}	not run								
	0	end	1.00×10^{13}	15.47 M	1.23 M	201.08 M	–	–	–	52.3
10^{-12}	10^{-40}	end	1.00×10^{13}	20.56 M	859.5 k	267.22 M	–	–	–	62.8
	10^{-150}	end	1.00×10^{13}	24.05 M	1.29 M	312.67 M	–	–	–	71.5
	10^{-300}	end	1.00×10^{13}	26.09 M	1.34 M	339.15 M	–	–	–	70.3
	0	end	1.00×10^{13}	26.09 M	1.34 M	339.15 M	–	–	–	82.0